ETSI TS 137 571-4 V19.0.0 (2026-08)



TECHNICAL SPECIFICATION

Universal Mobile Telecommunications System (UMTS);
LTE;
5G;
User Equipment (UE)
conformance specification for UE positioning;
Part 4: Test suites
(3GPP TS 37.571-4 version 19.0.0 Release 19)



Reference
RTS/TSGR-0537571-4vj00

Keywords
5G,LTE,UMTS

ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° w061004871

Important notice

The present document can be downloaded from the
ETSI Search & Browse Standards application.

The present document may be made available in electronic versions and/or in print. The content of any electronic and/or print versions of the present document shall not be modified without the prior written authorization of ETSI. In case of any existing or perceived difference in contents between such versions and/or in print, the prevailing version of an ETSI deliverable is the one made publicly available in PDF format on ETSI deliver repository.

Users should be aware that the present document may be revised or have its status changed, this information is available in the Milestones listing.

If you find errors in the present document, please send your comments to the relevant service listed under Committee Support Staff.

If you find a security vulnerability in the present document, please report it through our Coordinated Vulnerability Disclosure (CVD) program.

Notice of disclaimer & limitation of liability

The information provided in the present deliverable is directed solely to professionals who have the appropriate degree of experience to understand and interpret its content in accordance with generally accepted engineering or other professional standard and applicable regulations.
No recommendation as to products and services or vendors is made or should be implied.
No representation or warranty is made that this deliverable is technically accurate or sufficient or conforms to any law and/or governmental rule and/or regulation and further, no representation or warranty is made of merchantability or fitness for any particular purpose or against infringement of intellectual property rights.
In no event shall ETSI be held liable for loss of profits or any other incidental or consequential damages.

Any software contained in this deliverable is provided "AS IS" with no warranties, express or implied, including but not limited to, the warranties of merchantability, fitness for a particular purpose and non-infringement of intellectual property rights and ETSI shall not be held liable in any event for any damages whatsoever (including, without limitation, damages for loss of profits, business interruption, loss of information, or any other pecuniary loss) arising out of or related to the use of or inability to use the software.

Copyright Notification

No part of this document may be reproduced in any form, by any means and in any media, without the prior written authorization of ETSI and except as expressly permitted below.
By way of exception and when the document is a normative deliverable (European Standard (EN), Technical Specification (TS), Group Specification (GS) or ETSI Standard (ES)), ETSI authorizes to reproduce and incorporate into products, services and technical documentation only those extracts (e.g. templates) that are strictly necessary for the technical implementation of the normative deliverable, to ensure compliance with the latter.
Nothing in this notice shall be construed as limiting any mandatory exceptions to copyright provided by applicable law.

© ETSI 2026.
All rights reserved.

Intellectual Property Rights

Essential patents

IPRs essential or potentially essential to normative deliverables (European Standard (EN), Technical Specification (TS), Group Specification (GS) or ETSI Standard (ES)) may have been declared to ETSI. The declarations pertaining to these essential IPRs, if any, are publicly available for **ETSI members and non-members**, and can be found in ETSI SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available from the ETSI Secretariat. Latest updates are available on the ETSI IPR online database.

Pursuant to the ETSI Directives including the ETSI IPR Policy, no investigation regarding the essentiality of IPRs, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in ETSI SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

Trademarks

The present document may include trademarks and/or tradenames which are asserted and/or registered by their owners. ETSI claims no ownership of these except for any which are indicated as being the property of ETSI, and conveys no right to use or reproduce any trademark and/or tradename. Mention of those trademarks in the present document does not constitute an endorsement by ETSI of products, services or organizations associated with those trademarks.

DECT™, **PLUGTESTS™**, **UMTS™** and the ETSI logo are trademarks of ETSI registered for the benefit of its Members. **3GPP™**, **LTE™** and **5G™** logo are trademarks of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners. **oneM2M™** logo is a trademark of ETSI registered for the benefit of its Members and of the oneM2M Partners. **GSM**® and the GSM logo are trademarks registered and owned by the GSM Association.

Legal Notice

This Technical Specification (TS) has been produced by ETSI 3rd Generation Partnership Project (3GPP).

The present document may refer to technical specifications or reports using their 3GPP identities. These shall be interpreted as being references to the corresponding ETSI deliverables.

The cross reference between 3GPP and ETSI identities can be found at 3GPP to ETSI numbering cross-referencing.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

Contents

Intellectual Property Rights 2
Legal Notice 2
Modal verbs terminology 2
Foreword 6
Introduction 6
1 Scope 7
2 References 7
3 Definitions, symbols and abbreviations 9
3.1 Definitions 9
3.2 Symbols 9
3.3 Abbreviations 9
4 A-GPS system architecture and test model 9
4.1 ATS and module structure 9
4.2 Upper tester 9
4.3 SV PCO 9
4.4 A-GPS Primitives 9
4.5 Specific test suite operation for A-GPS testing 9
5 A-GNSS and LTE positioning system architecture and test models 10
5.1 Test system architecture 10
5.1.1 General system architecture 10
5.1.2 Component architecture 10
5.1.2.1 Interfaces 11
5.2 Test model 11
5.2.1 LTE GNSS test model 12
5.2.2 UTRA A-GNSS test model 13
5.3 ASP specifications 14
5.4 LTE OTDOA test model 14
5.5 LTE ECID test model 15
6 LTE indoor positioning system architecture and test models 15
6.1 Test system architecture 15
6.1.1 General system architecture 15
6.1.2 Component architecture 15
6.1.2.1 Interfaces 16
6.2 Test model 16
6.2.1 LTE indoor positioning test model 16
6A LTE test models requiring GNSS support 17
6B NG-RAN NR positioning system architecture and test models 17
6B.1 Test system architecture 17
6B.1.1 General system architecture 17
6B.1.2 Component architecture 17
6B.2 Test models 18
6B.2.1 NR GNSS positioning test model 18
6B.2.2 NR indoor positioning test model 19
6B.2.3 NR OTDOA (LTE) positioning test model 19
6B.2.4 NR Multi-RTT, DL-AoD and DL-TDOA positioning test model 21
6B.2.5 NR E-CID test model 21
6B.2.6 NR SLPP over NR sidelink test model 22
6C NR/5GC test models requiring GNSS support 23
6D Test methods and design considerations 24

6D.1 SLPP 24

7 Upper Tester Interface 25
7.1 MMI for A-GPS 25
7.2 MMI for A-GNSS, LTE and NR positioning 25

8 IXIT Proforma 25
8.1 A-GPS test suite parameters declarations 25
8.2 A-GNSS, LTE and NR positioning test suite parameters declarations 26

9 Other SS Requirements with TTCN-3 impact 26
9.1 Codec Requirements 26

Annex A (normative): Test Suites 27

A.1 UTRA A-GPS test suite 27
A.1.1 Baseline of specifications 27
A.1.2 AGPS ATS 27
A.1.3 The TTCN Machine Processable form (TTCN.MP) 28

A.2 UTRA A-GNSS, LTE and NR/5GC positioning test suite 28
A.2.1 Baseline of specifications 28
A.2.2 E-UTRA LTE positioning Test Cases 28
A.2.3 UTRA A-GNSS Test Cases 29
A.2.4 NR/5GC Test Cases 29

Annex B (informative): ASN.1 modules 31

B.1 A-GPS ASN.1 module 31

B.2 A-GNSS ASN.1 modules 31

B.3 LPP ASN.1 31

B.4 SLPP ASN.1 31

Annex C (informative) E-UTRA System Simulator Extension ASP Definitions in TTCN-3 32

C.1 EUTRA_POS_ASP_TypeDefs 32
C.1.1 System_Configuration_PosExt 32
C.1.2 Cell_Configuration_PosExt 32
C.1.2.1 Cell_Configuration_Common_PosExt 32
C.1.2.2 Downlink_Physical_Layer_Configuration_PosExt 33
C.1.3 System_Interface_PosExt 33

C.2 References to TTCN-3 34

Annex D (informative) Positioning System Simulator ASP Definitions in TTCN-3 35

D.1 PosSystem_ASP_TypeDefs 35
D.1.1 PosSystem_Load_Scenario 35
D.1.2 PosSystem_Retrieve_Data 36
D.1.2.1 SatSystem_Retrieve_UTRAN_Data 36
D.1.2.2 PosSystem_Retrieve_LPP_Data 37
D.1.2.3 PosSystem_Retrieve_Data_Common 38
D.1.3 PosSystem_PowerControl 38
D.1.4 PosSystem_GnssUtcTime 39
D.1.5 System_Interface 39

D.2 CommonDefs 40

D.3 References to TTCN-3 41

Annex E (informative) NR5GC System Simulator Extension ASP Definitions in TTCN-3 42

E.1 NR5GC_POS_ASP_TypeDefs 42
E.1.1 NR_System_Configuration_PosExt 42
E.1.2 NR_Cell_Configuration_PosExt 42

E.1.2.1 NR_Cell_Configuration_Common_PosExt 42
E.1.2.2 Downlink_Physical_Layer_Configuration_PosExt 43
E.1.3 NR_System_Interface_PosExt 44
E.2 References to TTCN-3 45
Annex F (informative) NR Sidelink System Simulator Extension ASP Definitions in TTCN-3 46
F.1 NR5GC_SL_POS_ASP_TypeDefs 46
F.1.1 NR_SL_System_Configuration_PosExt 46
F.1.2 NR_SL_SS_UE_Config_PosExt 46
F.1.2.1 NR_SL_PRS_Pool_Configuration 48
F.1.3 NR_Sidelink_System_Interface_PosExt 52
F.2 References to TTCN-3 52
Annex G (informative): Change history 53
History 61

Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

x the first digit:

1 presented to TSG for information;

2 presented to TSG for approval;

3 or greater indicates TSG approved document under change control.

y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

z the third digit is incremented when editorial only changes have been incorporated in the document.

Introduction

The present document is part 4 of a multi-part conformance test specification. The specification contains a TTCN design frame work and the detailed test specifications in TTCN for the UE:

- A-GPS at the UTRA Uu interface in TTCN-2;
- A-GNSS at the UTRA Uu interface in TTCN-3;
- LTE positioning at the LTE-Uu interface in TTCN-3.
- NR positioning at the NR-Uu interface in TTCN-3.

The present document is part 4 of a multi-parts TS:

- 3GPP TS 37.571-1: User Equipment (UE) conformance specification for UE positioning; Part 1: Terminal conformance.
- 3GPP TS 37.571-2: User Equipment (UE) conformance specification for UE positioning; Part 2: Protocol conformance.
- 3GPP TS 37.571-3: User Equipment (UE) conformance specification for UE positioning; Part 3: Implementation Conformance Statement (ICS).
- **3GPP TS 37.571-4: User Equipment (UE) conformance specification for UE positioning; Part 4: Test suites.**
- 3GPP TS 37.571-5: User Equipment (UE) conformance specification for UE positioning; Part 5: Test scenarios and assistance data.

1 Scope

The present document specifies the protocol and signalling conformance testing in TTCN for the UE:

- A-GPS at the UTRA Uu interface;
- LTE positioning at the LTE-Uu interface;
- A-GNSS at the UTRA Uu interface.
- NR positioning at the NR-Uu interface.

The following TTCN test specification and design considerations can be found in the present document:

- Test system architecture;
- Test models and ASP definitions;
- Test methods and usage of communication ports definitions;
- Test configurations;
- Design principles and assumptions;
- TTCN styles and conventions;
- Partial PIXIT proforma;
- Test suites in TTCN-2 and TTCN-3;
- The Test Suites designed and implemented in the present document are based on the test specifications in prose in 3GPP TS 37.571-2 [3];
- The applicability of the individual test cases is specified in the test ICS proforma specification in 3GPP TS 37.571-3 [4].

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document in the same Release as the present document unless the context in which the reference is made suggests a different Release is relevant (information on the applicable release in a particular context can be found in e.g. test case title, description or applicability, message description or content).

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 37.571-1: " User Equipment (UE) conformance specification for UE positioning; Part 1: Conformance test specification".

[3] 3GPP TS 37.571-2: " User Equipment (UE) conformance specification for UE positioning; Part 2: Protocol conformance".

[4] 3GPP TS 37.571-3: " User Equipment (UE) conformance specification for UE positioning; Part 3: Implementation Conformance Statement (ICS)".

[5] 3GPP TS 37.571-5: " User Equipment (UE) conformance specification for UE positioning; Part 5: Test scenarios and assistance data".

[6] 3GPP TS 36.523-3: "Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); User Equipment (UE) conformance specification; Part 3: Test suites".

[7] 3GPP TS 36.508: "Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); Common test environments for User Equipment (UE) conformance testing".

[8] 3GPP TS 34.123-3: "User Equipment (UE) conformance specification; Part 3: Abstract test suite (ATS)".

[9] 3GPP TS 34.108: "Common test environments for User Equipment (UE); Conformance testing".

[10] 3GPP TS 37.355: " LTE Positioning Protocol (LPP)".

[11] 3GPP TS 24.301: "Non-Access-Stratum (NAS) protocol for Evolved Packet System (EPS); Stage 3".

[12] 3GPP TS 25.331: "Radio Resource Control (RRC); Protocol specification".

[13] 3GPP TS 24.080: "Mobile radio interface layer 3 supplementary services specification; Formats and coding".

[14] 3GPP TS 29.002: "Mobile Application Part (MAP) specification".

[15] ITU-T Recommendation X.880: "Information technology - Remote Operations: Concepts, model and notation".

[16] ITU-T Recommendation Q.773: "Transaction capabilities formats and encoding".

[17] ETSI ES 201 873-1: "Methods for Testing and Specification (MTS); The Testing and Test Control Notation version 3; Part 1: TTCN-3 Core Language".

[18] 3GPP TS 34.109: "Terminal logical test interface; Special conformance testing functions".

[19] 3GPP TS 23.038: "Alphabets and language-specific information".

[20] ETSI TR 101 666 (V1.0.0): "Information technology; Open Systems Interconnection Conformance testing methodology and framework; The Tree and Tabular Combined Notation (TTCN) (Ed. 2++)".

[21] 3GPP TS 24.008: "Mobile radio interface Layer 3 specification; Core network protocols; Stage 3".

[22] 3GPP TS 23.032: "Universal Geographical Area Description (GAD)".

[23] 3GPP TS 36.211: "Evolved Universal Terrestrial Radio Access (E-UTRA); Physical channels and modulation".

[24] 3GPP TS 38.523-3: "5GS; User Equipment (UE) conformance specification; Part 3: Protocol Test Suites".

[25] 3GPP TS 24.501: "Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3".

[26] 3GPP TS 36.331: "Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification".

[27] 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification".

[28] 3GPP TS 38.508-1: "5GS; User Equipment (UE) conformance specification; Part 1: Common test environment".

[29] 3GPP TS 38.211: "NR; Physical channels and modulation".

[30] 3GPP TS 38.355: "NR; Sidelink Positioning Protocol (SLPP); Protocol specification".

[31] 3GPP TS 24.587: "Vehicle-to-Everything (V2X) services in 5G System (5GS); Stage 3".

3 Definitions, symbols and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] apply.

3.2 Symbols

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] apply.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply:

ASP Abstract Service Primitive

4 A-GPS system architecture and test model

A-GPS signalling conformance tests are specified in TTCN-2. The system architecture and the test model are based on 3GPP TS 34.123-3 [8].

4.1 ATS and module structure

Refer to 3GPP TS 34.123-3 [8], clause 5.

4.2 Upper tester

Refer to 3GPP TS 34.123-3 [8], clause 7.3.5.1.

4.3 SV PCO

Refer to 3GPP TS 34.123-3 [8], clause 7.3.5.2.

4.4 A-GPS Primitives

Refer to 3GPP TS 34.123-3[8], clause 7.3.5.3.

4.5 Specific test suite operation for A-GPS testing

Refer to 3GPP TS 34.123-3 [8], clause 8.7.7.

5 A-GNSS and LTE positioning system architecture and test models

5.1 Test system architecture

5.1.1 General system architecture

Refer to 3GPP TS 36.523-3 [6], clause 4.1.1.

5.1.2 Component architecture

Each access technology (RAT) is hosted by a separate TTCN-3 parallel component (PTC):

- E-UTRA PTC;
- UTRA PTC.

The PTCs are controlled by the TTCN-3 positioning master test component (POS MTC) which:

- Is an extension of the MTC as defined in 3GPP TS 36.523-3 [6], clause 4.1.2.
- Controls the SS Positioning Simulator.

Figure 5.1.2-1 shows this component architecture for an E-UTRA and UTRA scenario.

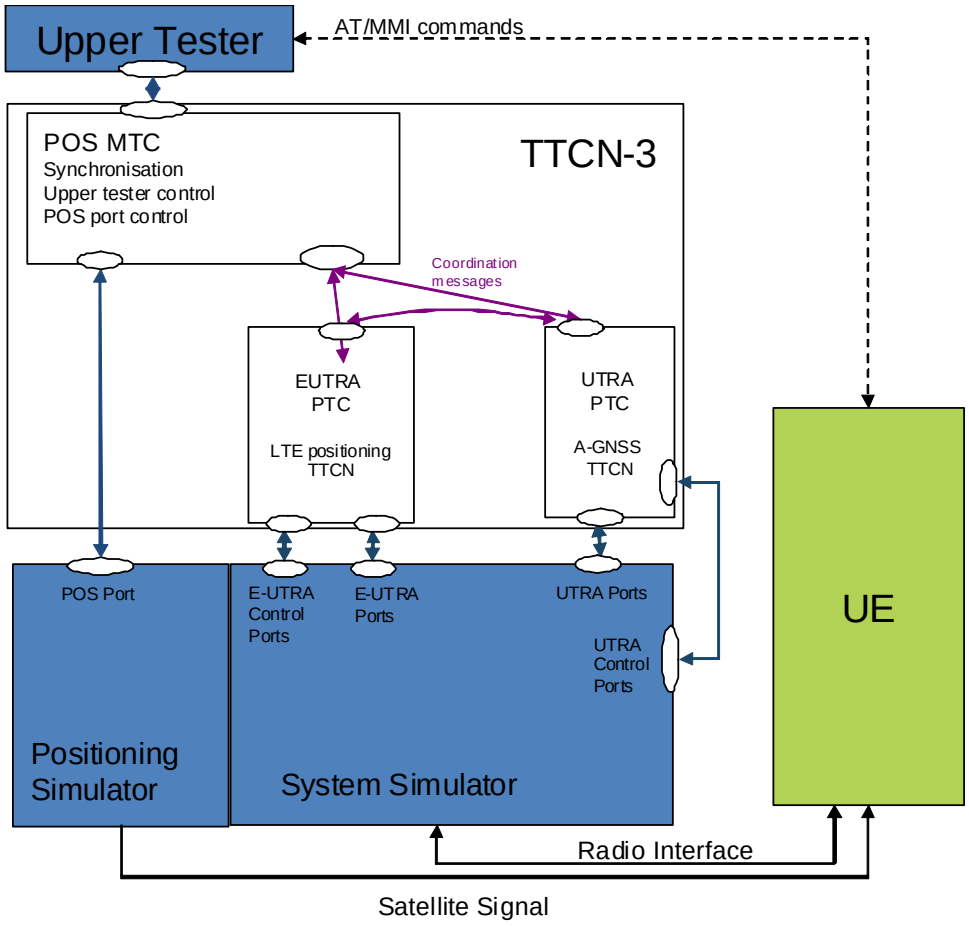


Figure 5.1.2-1: Component architecture for an E-UTRA and UTRA scenario

The PTCs are allocated according to the following principles:

- For LTE positioning tests cases, the E-UTRA PTC is created.
- For LTE positioning CS fallback test cases, E-UTRA and UTRA PTCs are created.
- For UTRA A-GNSS test cases, the UTRA PTC is created.

5.1.2.1 Interfaces

A number of interfaces are taken into consideration.

POS MTC - PTC:

Common synchronisation of PTCs.

Positioning simulator command/data coordination messages.

Upper tester primitives.

POS MTC - System Interface:

Upper tester primitives.

POS MTC- Positioning Simulator:

Positioning simulator command and data primitives.

PTC - PTC:

Primitives containing information for IRAT handover.

PTC - System Interface:

Primitives containing peer-to-peer message.

Configuration primitives.

5.2 Test model

In order to perform the LTE positioning and A-GNSS tests, the Positioning Simulator shall have two basic functional unites:

- Satellite simulator generating and broadcasting satellite signals.
- Assistance data source storing the data associated with the pre-defined GPS and GNSS test scenarios (see 3GPP TS 37.571-5 [5]).

This specification supports the following GNSS: GPS, Galileo, GLONASS and BDS.

5.2.1 LTE GNSS test model

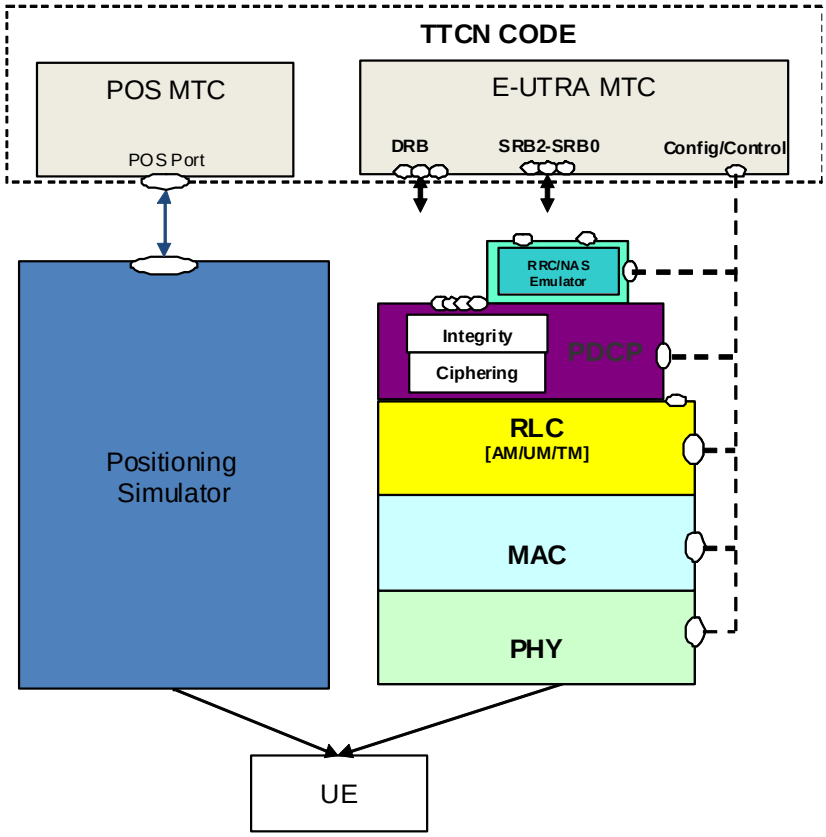


Figure 5.2.1-1: E-UTRA Test model for LTE positioning GNSS testing

The LTE positioning test model consists of an SS E-UTRAN protocol stack and a positioning simulator. The SS E-UTRA emulation part is the same as the model defined in 3GPP TS 36.523-3 [6].

The UE is configured in normal mode. On UE side Ciphering/Integrity (PDCP and NAS) is enabled. The tests are performed on C-Plane.

The ASPs at POS Port are used to:

- Load a pre-defined test scenario into the positioning simulator.
- Start or stop generating and broadcasting satellite signals from the positioning simulator.
- Retrieve the GNSS assistance data from assistance data source.

Under the ASP commands, the upper tester loads or re-loads pre-defined GNSS test scenario to the positioning simulator. The generated satellite signals shall simulate the GNSS environment as specified in 3GPP TS 37.571-2 [3], clause 5.

The SS also sends the GNSS assistance data to the UE through LPP signalling (see 3GPP TS 37.355 [10]) to facilitate the UE acquiring and tracking satellites. Such assistance data shall be consistent to within ±2 seconds with the satellite signals generated (see 3GPP TS 37.571-2 [3], clause 5.2.1).

The assistance data source shall provide the assistance data consistent to +1/-0 second with the GNSS test scenario currently running in the satellite simulator (i.e. the data shall be up to 1 second in advance of the scenario); this allows for a further 2 seconds of latency in the SS. In case multiple satellite systems are configured in a test, the assistance data are requested or retrieved in a single ASP command for all satellite systems.

The UE positioning services are encoded in a TCAP [16] ROSE component [15] of the facility information element [14] in the Supplement Service [13] conveyed in the NAS messages. The facility information element is encoded in BER in TTCN referred to 3GPP TS 24.080 [13], clause 3.6.1 and 3GPP TS 29.002 [14], clause 17.1.1.

5.2.2 UTRA A-GNSS test model

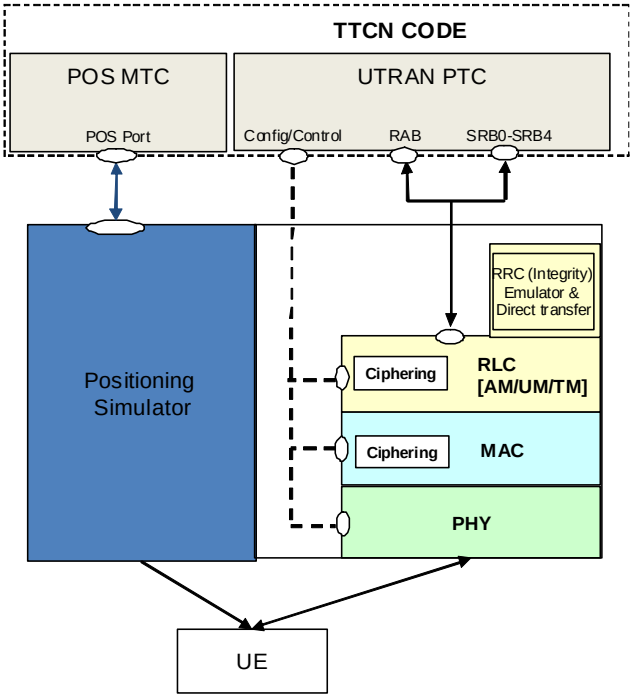


Figure 5.2.2-1: UTRA Test model for A-GNSS testing

The A-GNSS test model consists of an SS UTRAN protocol stack and a positioning simulator. The SS UTRA emulation part is the same as the model defined in 3GPP TS 36.523-3 [6], clause 4.4.1.

The UE is configured in normal mode. Ciphering and Integrity (RRC) are enabled. The tests are performed on C-Plane.

The ASPs at POS Port are used to:

- Load a pre-defined test scenario into the positioning simulator.
- Start or stop generating and broadcasting satellite signals from the positioning simulator.
- Retrieve the GPS and GNSS assistance data from assistance data source.

Under the ASP commands, the upper tester loads or re-loads pre-defined GPS or GNSS test scenario to the positioning simulator. The generated satellite signals shall simulate the A-GPS and A-GNSS environment as specified in 3GPP TS 37.571-2 [3], clause 4.2.

The SS also sends the GPS or GNSS assistance data to the UE through RRC signalling to facilitate the UE acquiring and tracking satellites. Such assistance data shall be consistent to within ±2 seconds with the satellite signals generated (see 3GPP TS 37.571-2 [3], clause 4.2).

The assistance data source shall provide the assistance data consistent to +1/-0 second with the GPS or GNSS test scenario currently running in the satellite simulator (i.e. the data shall be up to 1 second in advance of the scenario); this allows for a further 2 seconds of latency in the SS. In case multiple satellite systems are configured in a test, the assistance data are requested or retrieved in a single ASP command for all satellite systems.

The UE positioning services are encoded in a TCAP [16] ROSE component [15] of the facility information element [14] in the Supplement Service [13] conveyed in the NAS messages. The facility information element is encoded in BER in TTCN referred to 3GPP TS 24.080 [13], clause 3.6.1 and 3GPP TS 29.002 [14], clause 17.1.1.

5.3 ASP specifications

The E-UTRAN and UTRAN ASP definitions are specified according to 3GPP TS 36.523-3 [6], clause 6.

The positioning simulator ASP definitions are documented in annex C.

5.4 LTE OTDOA test model

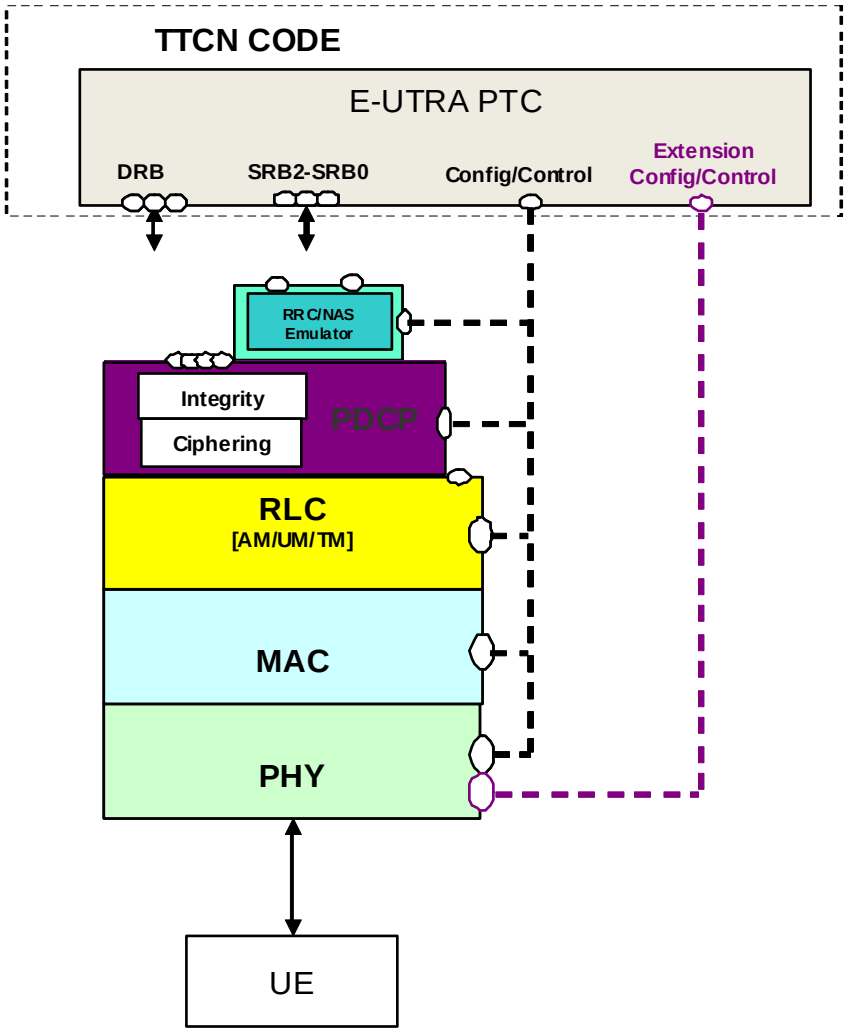


Figure 5.4-1: E-UTRA OTDOA test model

The LTE OTDOA test model consists of an SS E-UTRAN protocol stack. The SS E-UTRA emulation part is the same as the model defined in 3GPP TS 36.523-3 [6] with an additional port to configure the antenna port 6 and the Positioning Reference Signal (PRS) in the LTE cells. The extension of LTE ASPs is defined in annex C.

The simulated OTDOA environment is specified in 3GPP TS 37.571-2 [3], clause 5.2.2), cells are synchronous and configured with the timing offset set to 0. The PRS is transmitted according to 3GPP TS 36.211 [23], clause 6.10.4.

The UE is configured in normal mode. On UE side Ciphering/Integrity (PDCP and NAS) is enabled. The tests are performed on C-Plane.

The PRS signals are transmitted by the SS E-UTRAN, the assistance data are written in TTCN.

5.5 LTE ECID test model

The test model is referred to the LTE positioning test model defined in clause 5.2.1; except that no satellite signals are required.

6 LTE indoor positioning system architecture and test models

6.1 Test system architecture

6.1.1 General system architecture

Refer to 3GPP TS 36.523-3 [6], clause 4.1.1.

6.1.2 Component architecture

The same architecture as in 5.1.2 applies, excluding the UTRA PTC.

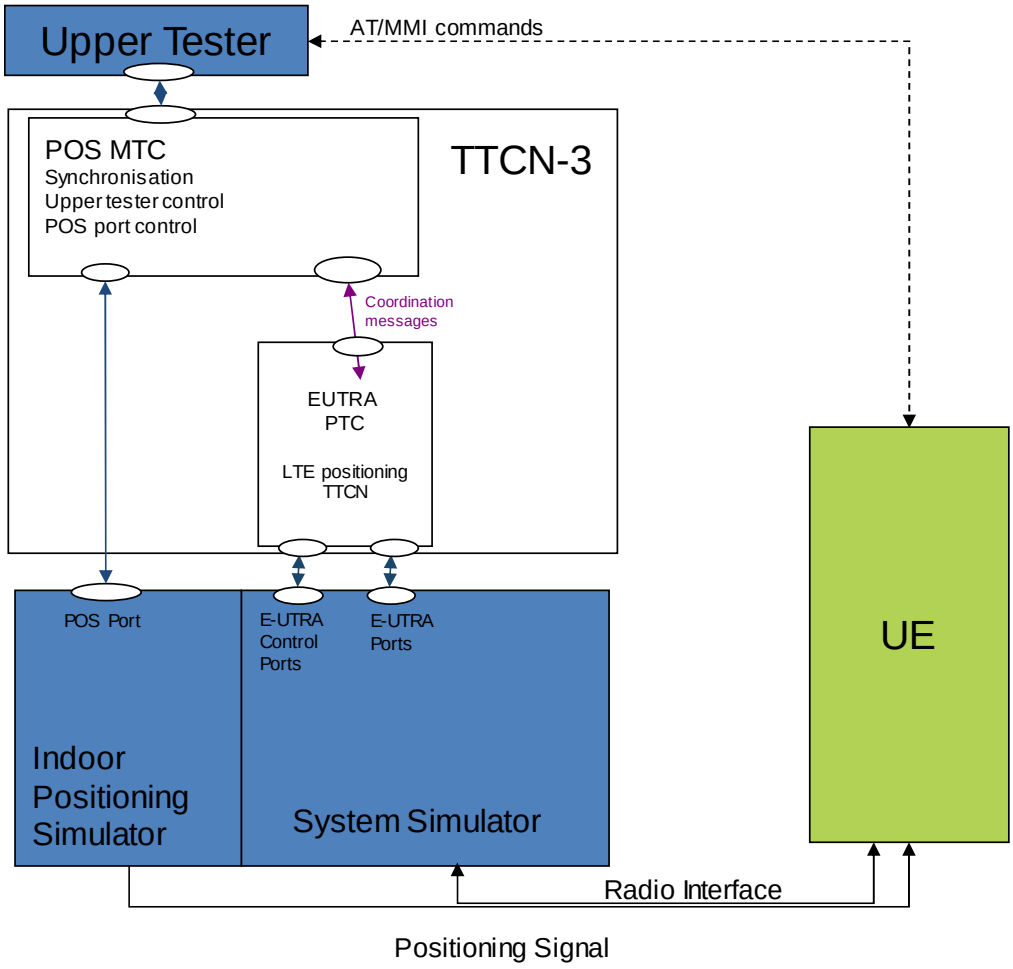


Figure 6.1.2-1: Component architecture for an E-UTRA scenario

6.1.2.1 Interfaces

Refer to 5.1.2.1.

6.2 Test model

6.2.1 LTE indoor positioning test model

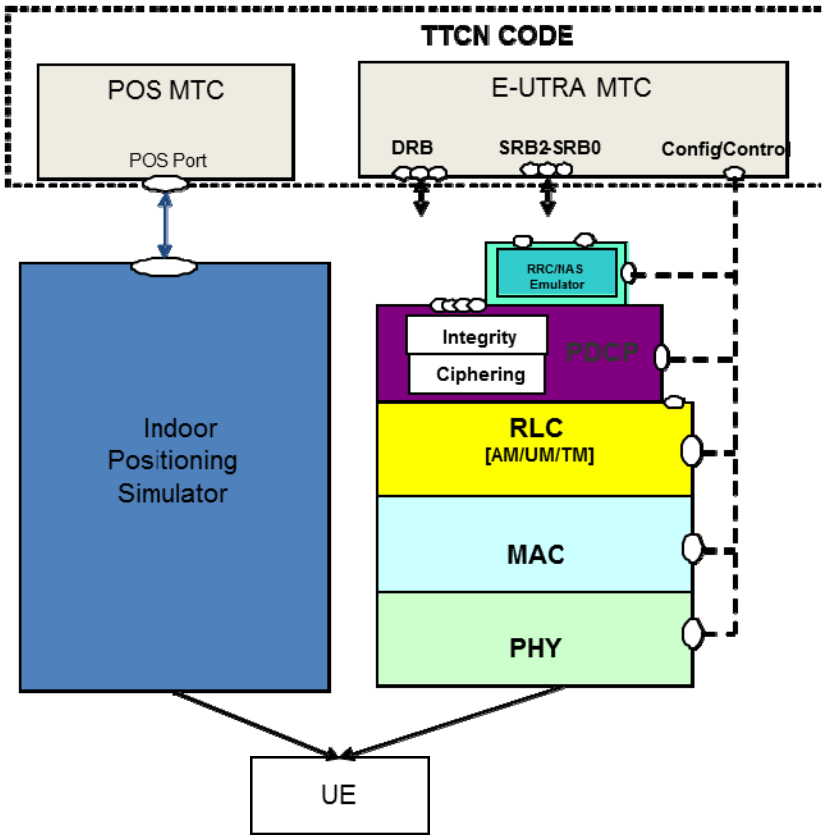


Figure 6.2.1-1: E-UTRA Test model for LTE indoor positioning testing

The LTE Indoor Positioning test model consists of an SS E-UTRAN protocol stack and an Indoor Positioning simulator. Depending on the Indoor Positioning Technology to be tested, the Indoor Positioning simulator will generate:

- MBS beacon signals, for MBS test cases;
- WLAN signals, for WLAN test cases;
- Bluetooth signals, for Bluetooth test cases.
- The Indoor Positioning simulator will be switched off for Sensor test cases.

The SS E-UTRA emulation part is the same as the model defined in 3GPP TS 36.523-3 [6].

The UE is configured in normal mode. On UE side Ciphering/Integrity (PDCP and NAS) is enabled. The tests are performed on C-Plane.

The ASPs at POS Port are used to:

- Load a pre-defined test scenario into the Indoor Positioning simulator.
- Start or stop generating and broadcasting signals from the Indoor Positioning simulator.

- Retrieve the appropriate (MBS, WLAN, Sensor) assistance data from the assistance data source.

Under the ASP commands, the upper tester loads or re-loads pre-defined test scenario to the Indoor Positioning simulator. The generated signals shall simulate the corresponding Indoor Positioning environment as specified in 3GPP TS 37.571-2 [3], clause 5.

The SS also sends the appropriate assistance data to the UE through LPP signalling (see 3GPP TS 37.355 [10]) to facilitate the UE acquisition of MBS Beacons or WLAN APs. For UE-based tests, the SS also sends the appropriate assistance data to enable the UE to calculate its position for MBS or WLAN tests, or calculate altitude based on barometric pressure sensor measurements.

The UE positioning services are encoded in a TCAP [16] ROSE component [15] of the facility information element [14] in the Supplement Service [13] conveyed in the NAS messages. The facility information element is encoded in BER in TTCN referred to 3GPP TS 24.080 [13], clause 3.6.1 and 3GPP TS 29.002 [14], clause 17.1.1.

6A LTE test models requiring GNSS support

Several LTE test models specified in TS 36.523-3 [6] require the presence of a GNSS emulation. Those are:

- V2X services: refer to TS 36.523-3 [6] clause 4.2.11
- Aerial UE: refer to TS 36.523-3 [6] clause 4.2.12

The positioning simulator shall be capable of generating and broadcasting the following GNSS: GPS, Galileo, GLONASS and BDS.

The ASPs at POS Port for the both test models are used to:

- Load a pre-defined test scenario into the positioning simulator.
- Start or stop generating and broadcasting satellite signals from the positioning simulator.

The ASPs at POS Port for V2X services are additionally used to:

- Trigger the positioning simulator to move in the test scenario to the next geographical area or zone.
- Trigger the positioning simulator to switch on/off the transmitter without stopping the UTC time sequence.

The ASPs at POS Port for aerial UE communication are additionally used to:

- Set an initial altitude above the sea level. Geographical latitude and longitude are defined in the test scenario and remain constant during the test execution.
- Trigger the positioning simulator to simulate a vertical movement in the test scenario towards a specified altitude level.

6B NG-RAN NR positioning system architecture and test models

6B.1 Test system architecture

6B.1.1 General system architecture

Refer to 3GPP TS 36.523-3 [6], clause 4.1.1.

6B.1.2 Component architecture

The NR PTC is controlled by the TTCN-3 positioning master test component (POS MTC) which:

- Is an extension of the MTC as defined in 3GPP TS 38.523-3 [24], clause 4.2.
- Controls the SS Positioning Simulator.

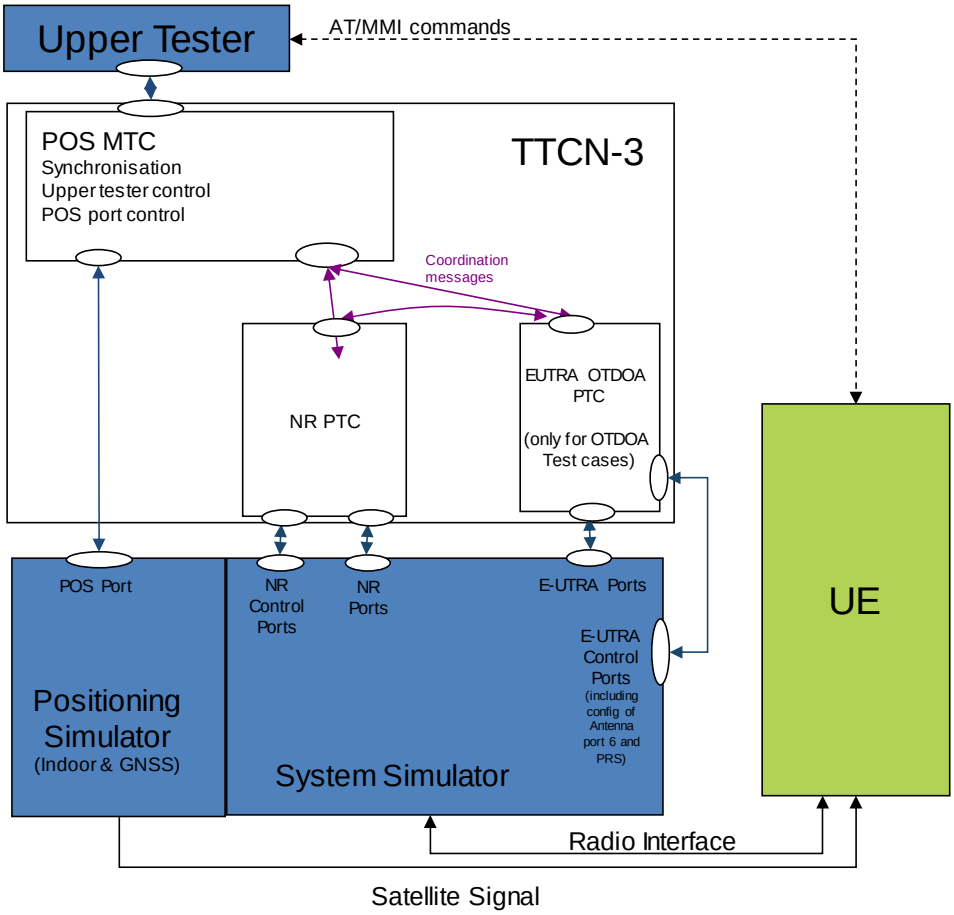


Figure 6B.1.2-1: Component architecture for a NR scenario

As specified in TS 37.571-2 [2], subclause 8.2.8:

- The connection between the Position Simulator and the UE (Satellite signal) shall be conducted.

- The connection between the System Simulator and the UE (Radio Interface) shall be conducted for NR FR1 and radiated (OTA) for NR FR2.

6B.2 Test models

6B.2.1 NR GNSS positioning test model

The NR positioning test model consists of an SS NR protocol stack and a positioning simulator. The SS NR/5GC emulation part is the same as the model defined in 3GPP TS 38.523-3 [24]. The positioning simulator is the same as for LTE positioning test model defined in clause 5.2.1.

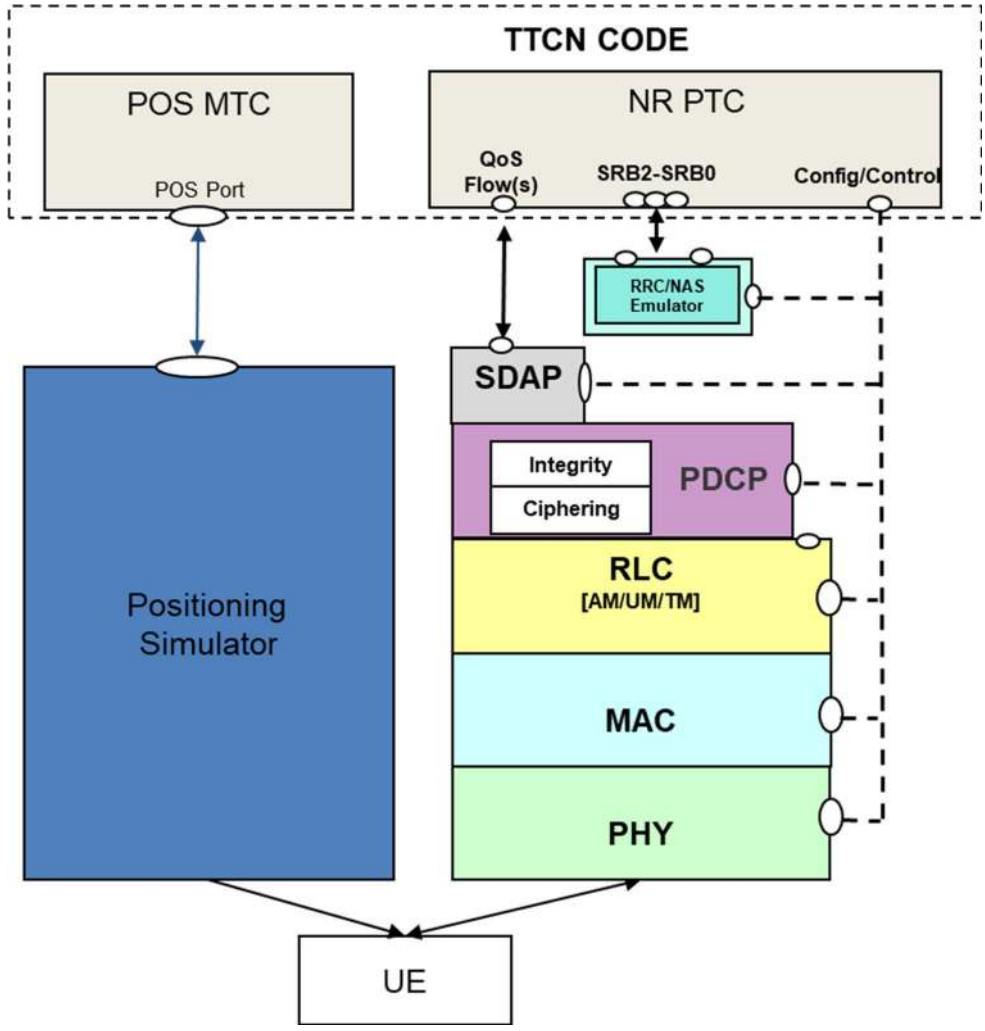


Figure 6B.2.1-1: Test model for NR GNSS positioning testing

6B.2.2 NR indoor positioning test model

The NR indoor positioning test model is the same as the NR positioning test model in clause 6B.2.1 but replacing the Positioning simulator with an Indoor Positioning simulator. The same principles as in clause 6.2.1 apply but replacing LTE with NR.

6B.2.3 NR OTDOA (LTE) positioning test model

OTDOA measurements are not supported on NR cells in Rel-15. The OTDOA measurements are provided by LTE cells instead. The UE will be attached to an NR cell and the LPP session will be established over NR/5GC NAS. The UE will report OTDOA RSTD measurements from LTE neighbour cells.

The NR OTDOA (LTE) positioning model consists of an SS NR/5GC protocol stack and a SS E-UTRAN protocol stack. The SS NR/5GC emulation part is the same as the model defined in 3GPP TS 38.523-3 [34]. The SS E-UTRAN emulation part is the same as the model defined in clause 5.4 (i.e. including an additional port to configure the antenna port 6 and the Positioning Reference Signal (PRS) in the LTE cells). The E-UTRA PTC will not be connected to NASEMU or IMS/IP PTCs.

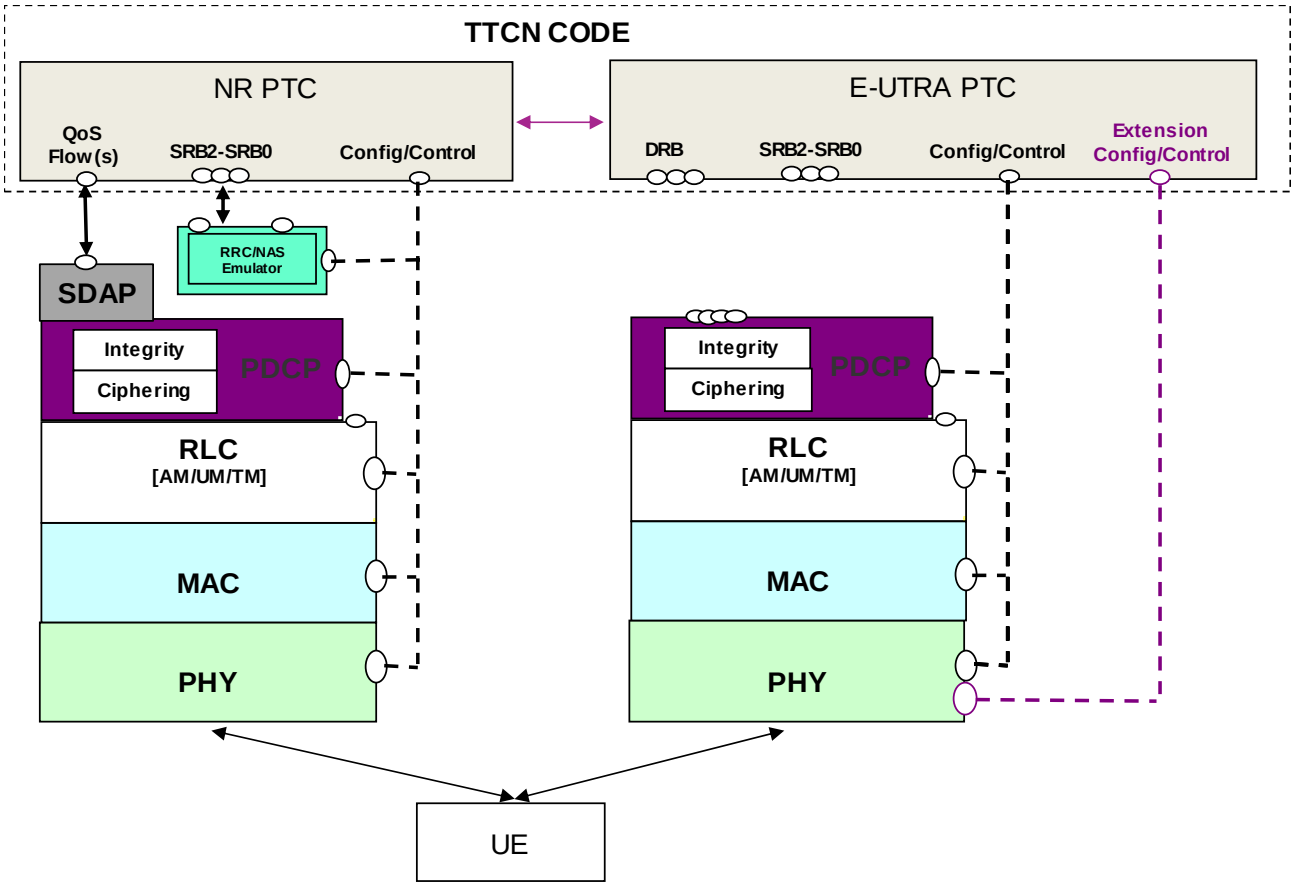


Figure 6B.2.3-1: Test model for NR OTDOA (LTE) positioning testing

6B.2.4 NR Multi-RTT, DL-AoD and DL-TDOA positioning test model

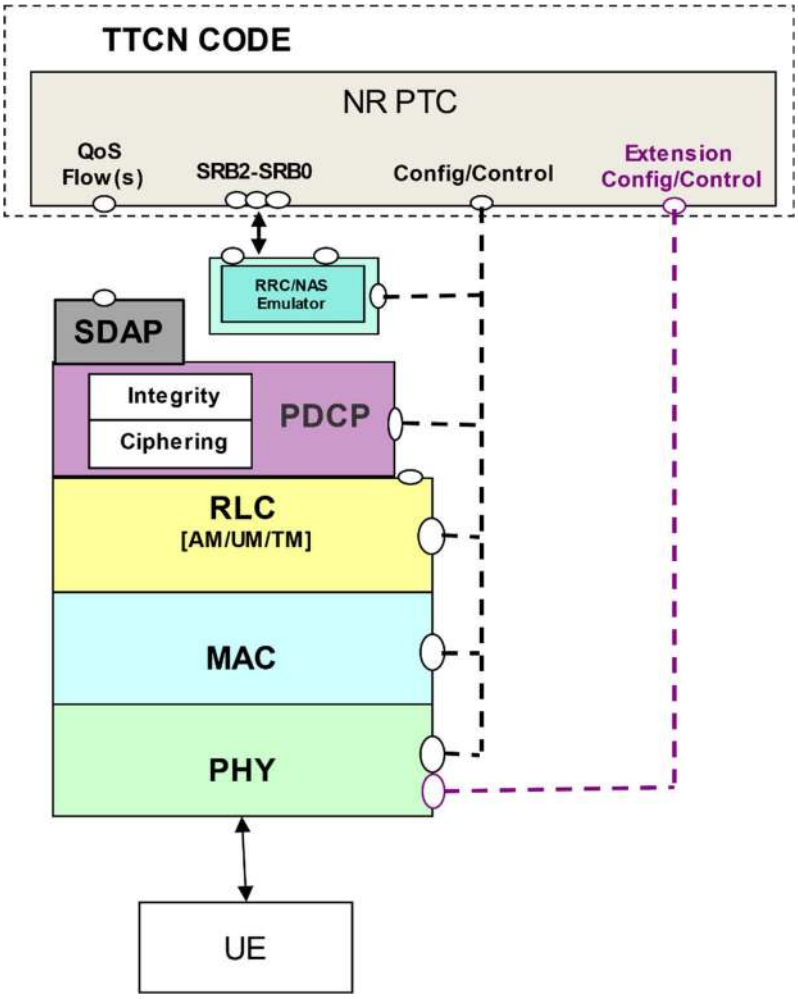


Figure 6B.2.4-1: NR Multi-RTT, DL-AoD and DL-TDOA test model

The NR Multi-RTT, DL-AoD and DL-TDOA test model is the same as the NR/5GC Layer 3 test model (TS 38.523-3 [24], clause 5.2.1.1) with an additional port to configure the DL Positioning Reference Signal (DL-PRS) in the NR cells. The extension of NR ASPs is defined in annex X [TBD].

The simulated NR Multi-RTT, DL-AoD and DL-TDOA environments are specified in TS 37.571-2 [3], clauses 8.2.9, 8.2.10 and 8.2.11, respectively. The DL-PRS is transmitted according to TS 38.211 [29], clause 7.4.1.7.

The UE is configured in normal mode. On UE side Ciphering/Integrity (PDCP and NAS) is enabled. The tests are performed on C-Plane.

The DL-PRS is transmitted by the physical layer of the NR cell(s) in the SS. The assistance data to provide the UE with the configuration of the DL-PRS is sent from the SS to the UE via LPP signalling.

For Multi-RTT, the SS will be also configured through the NR_SYS port to receive UL-SRS from the UE.

6B.2.5 NR E-CID test model

The test model is the same as the NR/5GC Layer 3 test model (TS 38.523-3 [24], clause 5.2.1.1).

6B.2.6 NR SLPP over NR sidelink test model

The NR SLPP over NR sidelink test model builds on top of the Layer 3 NR sidelink PC5 test models for unicast specified in TS 38.523-3 [24], clause 5.3.1, with the differences specified hereafter.

NOTE: In TS 37.571-2 [6] the PC5 unicast connection can be established using either V2X or 5G ProSe, depending on UE capabilities. The test model in this clause only considers the use of V2X. The test model with 5G ProSe is not supported in the present version of the specification.

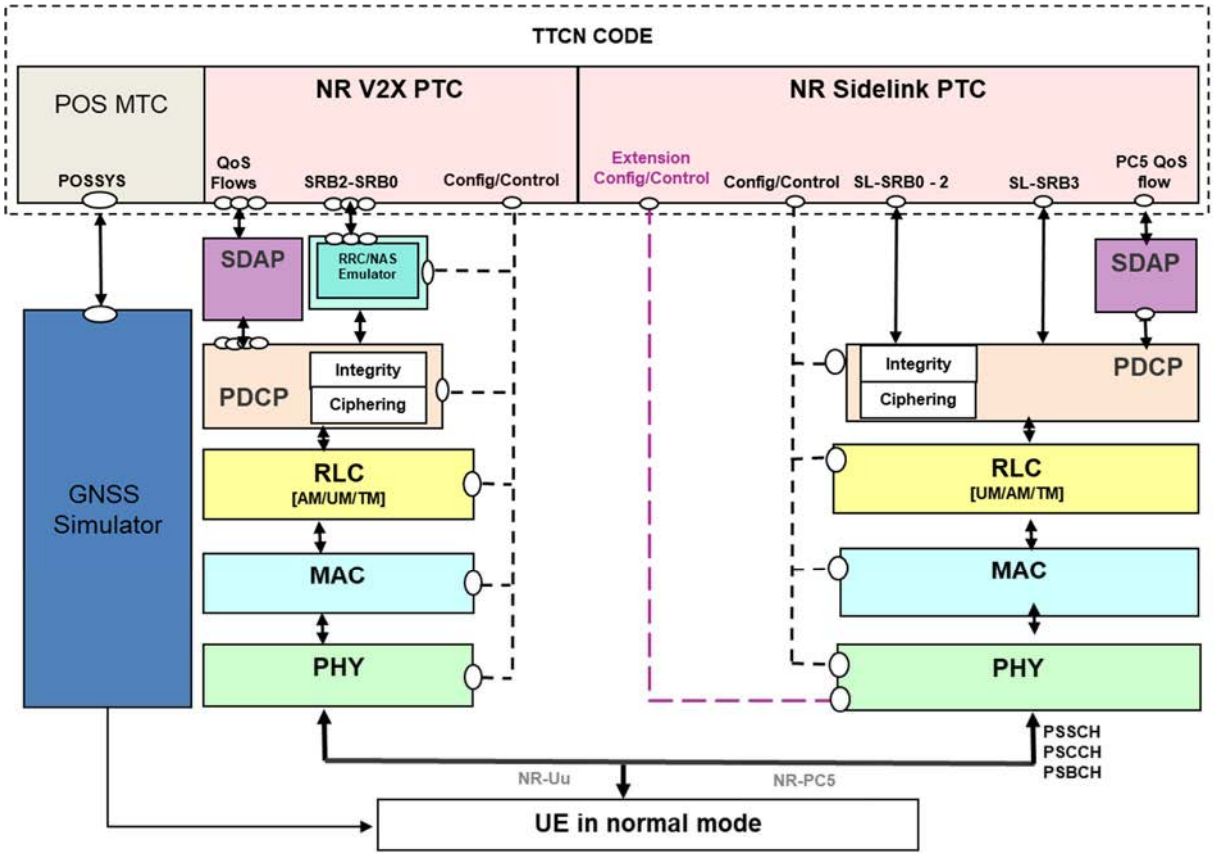


Figure 6B.2.6-1: Test model for NR SLPP positioning over NR sidelink in coverage

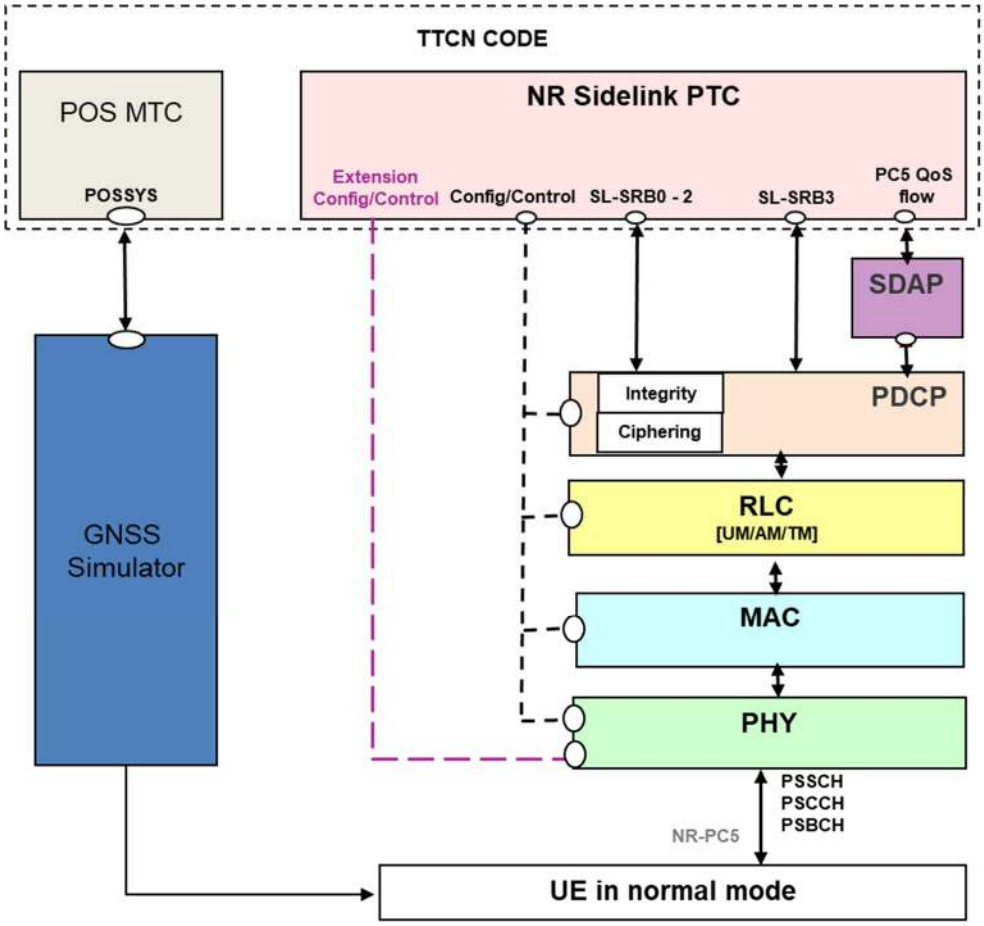


Figure 6B.2.6-2: Test model for NR SLPP positioning over NR sidelink out of coverage

The UE is configured in normal mode; i.e. Test Loop Mode E is not used.

When the UE is in coverage of an NR cell, this is depicted in the test model of Figure 6B.2.6-1. When the UE is out of coverage of an NR cell, this is depicted in the test model of Figure 6B.2.6-2. The NR-SS-UE can be configured to transmit SL-PRS on PC5.

In the NR-SS-UE, PDCP is configured in normal way with “Non-IP” when SLPP messages are being exchanged. The NR_SL_DATA port for transmission and reception of SLPP messages is above SDAP. SLPP messages are encoded/decoded according to TS 38.355 [30], clause 6.1. TTCN shall take care of adding (in transmission) / removing (in reception) the “Non-IP” header as described in TS 24.587 [31] clause 9.3.

The NR sidelink PTC has an additional port to configure SL Positioning Reference Signal (SL-PRS) in the NR-SS-UE(s) if these are required in the test case. The extension of the ASPs is defined in Annex F.

The simulated SLPP environment is specified in TS 37.571-2 [3], clauses 8.2.14 (for SL-RTT subtests), 8.2.15 (for SL-AoA subtests), 8.2.16 (for SL-TDOA subtests) and 8.2.17 (for SL-TOA subtests). The SL-PRS is transmitted according to TS 38.211 [29], clause 8.4.1.6. The assistance data to provide the UE with the configuration of the SL-PRS is sent from the NR-SS-UE to the UE via SLPP signalling.

6C NR/5GC test models requiring GNSS support

Several NR/5GC test models specified in TS 38.523-3 [24] require the presence of a GNSS emulation. Those are:

- NR sidelink: refer to TS 38.523-3 [24] clause 5.3.
- Uncrewed Aerial Vehicles (UAV): refer to TS 38.523-3 [24] clause 5.6.

The positioning simulator shall be capable of generating and broadcasting the following GNSS: GPS, Galileo, GLONASS and BDS.

The ASPs at POS Port for both test models are used to:

- Load a pre-defined test scenario into the positioning simulator.
- Start or stop generating and broadcasting satellite signals from the positioning simulator.

The ASPs at POS Port for NR sidelink are additionally used to:

- Trigger the positioning simulator to move in the test scenario to the next geographical area or zone.
- Trigger the positioning simulator to switch on/off the transmitter without stopping the UTC time sequence.
- Retrieve the current UTC time of the GNSS scenario.

The ASPs at POS Port for Uncrewed Aerial Vehicles are additionally used to:

- Set an initial altitude above the sea level. Geographical latitude and longitude are defined in the test scenario and remain constant during the test execution.
- Trigger the positioning simulator to simulate a vertical movement in the test scenario towards a specified altitude level.

6D Test methods and design considerations

6D.1 SLPP

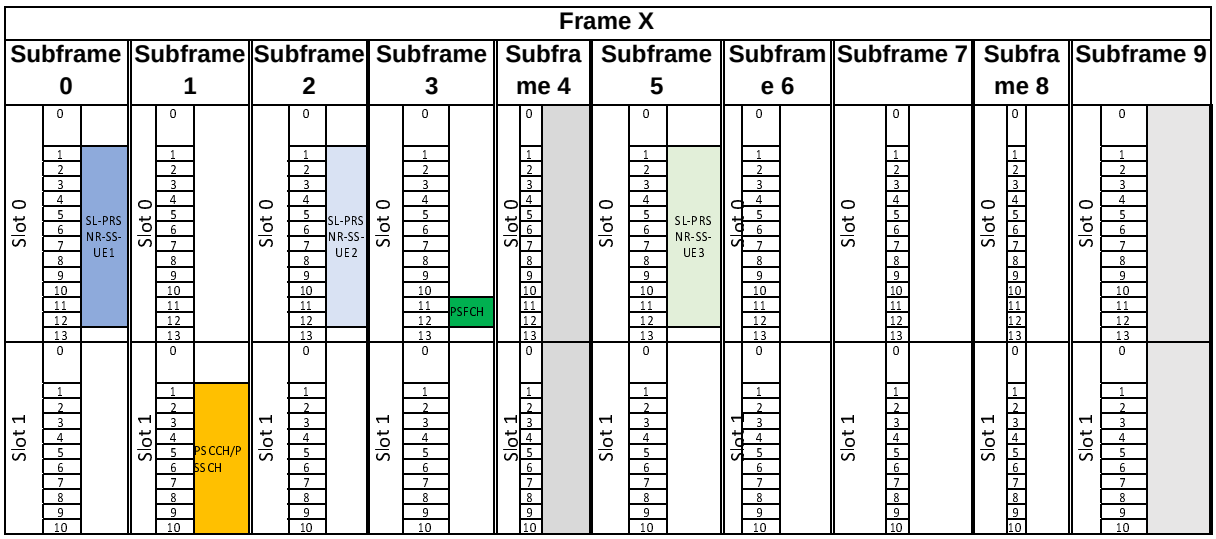
According to TS 38.508-1 [28], sl-TimeResource-r16 and sl-TimeResource-r18 are set to ‘1111111100’B.

When several (up to three) NR-SS-UE(s) are configured to transmit SL-PRS, TTCN configures each NR-SS-UE to transmit in a different subframe using the ASPs in Annex F (SL_TimeResource in NR_SL_PRS_SharedResourcePool_Type or NR_SL_PRS_DedicatedResourcePool_Type, depending on the case):

- NR-SS-UE1 is configured to transmit SL-PRS in slot 0,
- NR-SS-UE2 is configured to transmit SL-PRS in slot 4, and
- NR-SS-UE3 is configured to transmit SL-PRS in slot 10.

This is illustrated in Table 6D.1-1 below.

Table 6D.1-1: SL-PRS scheduling for FR1: SCS=30kHz





NOTE: Scheduling of PSCCH/PSSCH/PSFCH are shown for clarity, following TS 38.523-3 [24], Table 7.4.2.6-1.

7 Upper Tester Interface

7.1 MMI for A-GPS

Refer to 3GPP TS 34.123-3 [8], clause B.1.12.

7.2 MMI for A-GNSS, LTE and NR positioning

Refer to 3GPP TS 36.523-3 [6], clause 5.

The following MMI commands are defined for A-GNSS, LTE and NR positioning.

Table 7.2-1: MMI commands

Command	Parameters	
	Name	Value
"CHECK_LCS_CLIENT"	none	
“LOCATION_INFO”	none	

Table 7.2-2: MMI commands imported from 3GPP TS 36.523-3 [6], clause 5

Command
"SWITCH_ON"
"SWITCH_OFF"
"POWER_ON"
"POWER_OFF"
"CHECK_DTCH_THROUGHCONNECTED"
"CLEAR_STORED_ASSISTANCE_DATA"

The following AT commands are applied in TTCN.

Table 7.2-3: AT Commands

Command	Reference
AT+CMOLR	TS 27.007 [32]
AT+CMTLR	TS 27.007 [32]
AT+CMTLRA	TS 27.007 [32]

8 IXIT Proforma

8.1 A-GPS test suite parameters declarations

Refer to 3GPP TS 34.123-3 [8], clause B.1.13.

8.2 A-GNSS, LTE and NR positioning test suite parameters declarations

Same PIXIT parameters are referred to clause 8.1.

Table 8.2-1: Void

9 Other SS Requirements with TTCN-3 impact

9.1 Codec Requirements

Refer to 3GPP TS 36.523-3 [6] clause 8.1 and TS 38.523-3 [24] clause 8.1. In addition, the SS shall also comply with the codec requirements specified in Table 9.1-1.

Table 9.1-1: Codec requirements

Type definitions	Codec requirements	Encoding rule in TTCN-3
SLPP ASN.1 types used for SLPP signalling	shall comply to TS 38.355 [30] clause 6.1	UNALIGNED_PER_OctetAligned

Annex A (normative): Test Suites

This annex contains the approved TTCN Test Suites.

A.1 UTRA A-GPS test suite

This annex includes the reference to those approved A-GPS ATS which are contained in 3GPP TS 34.123-3 [8], clause A.11.

The ATSs have been produced using the Tree and Tabular Combined Notation (TTCN2) according to TR 101 666 [20].

A.1.1 Baseline of specifications

Table A.1.1-1 shows the baseline of the relevant core specifications and the test specifications which the delivered ATSs are referred to.

Table A.1.1-1: Versions of the test and Core specifications

Core specifications	3GPP TS 25.331 [12]
Test specifications	3GPP TS 37.571-2 [3]
	3GPP TS 37.571-3 [4]
	3GPP TS 37.571-5 [6]
	3GPP TS 34.123-3 [8]
	3GPP TS 34.108 [9]
	3GPP TS 34.109 [18]

A.1.2 AGPS ATS

The approved AGPS test cases are listed. An "X" in columns FDD or LCR TDD indicates the test case approved for the relevant ATS.

Table A.1.2-1: AGPS TTCN test cases

Test case reference		Description	FDD	TDD
TS 37.571-2 [3]	TS 34.123-3 [8]			
6.1.1.1	17.2.2.1	LCS Network Induced location request/UE-Based GPS/Emergency Call/with USIM	X	
6.1.1.2	17.2.2.2	LCS Network Induced location request/UE-Based GPS/Emergency Call/without USIM	X	
6.1.1.3	17.2.2.3	LCS Network induced location request/UE-Assisted GPS/Emergency call/With USIM	X	
6.1.1.4	17.2.2.4	LCS Network induced location request/UE-Assisted GPS/Emergency call/Without USIM	X	
6.1.2.1	17.2.3.2	LCS Mobile originated location request/UE-Based GPS/Position estimate request/Success	X	
6.1.2.7	17.2.3.9	LCS Mobile originated location request/UE-Assisted GPS/Position Estimate/Success	X	
6.1.3.1	17.2.4.1	LCS Mobile terminated location request/UE-Based GPS	X	
6.1.3.2	17.2.4.2	LCS Mobile-terminated location request/UE-Based GPS/Request for additional assistance data/Success	X	
6.1.3.3	17.2.4.3	LCS Mobile-terminated location request/UE-Based GPS/Failure - Not Enough Satellites	X	
6.1.3.4	17.2.4.4	LCS Mobile terminated location request/UE-Assisted GPS/Success	X	
6.1.3.5	17.2.4.5	LCS Mobile terminated location request/UE-Assisted GPS/Request for additional assistance data/Success	X	
6.1.3.7	17.2.4.7	LCS Mobile terminated location request/UE-Based GPS/Privacy Verification/Location Not Allowed if No Response	X	

A.1.3 The TTCN Machine Processable form (TTCN.MP)

The TTCN.MP representation corresponding to the corresponding FDD ATS is referred to 3GPP TS 34.123-3 [8], clause A.11.

A.2 UTRA A-GNSS, LTE and NR/5GC positioning test suite

The test suites have been produced using the Testing and Test Control Notation version 3 (TTCN-3) according to ES 201 873-1 [17].

A.2.1 Baseline of specifications

Table A.2.1-1 shows the baseline of the relevant core specifications and the test specifications which the delivered TTCN test suites are referred to.

Table A.2.1-1: References of the test and Core specifications

Core specifications baseline	3GPP TS 37.355 [10]
	3GPP TS 24.301 [11]
	3GPP TS 25.331 [12]
	3GPP TS 24.501 [25]
	3GPP TS 36.331 [26]
	3GPP TS 38.331 [27]
Test specifications	3GPP TS 37.571-2 [3]
	3GPP TS 37.571-3 [4]
	3GPP TS 37.571-5 [5]
	3GPP TS 36.508 [7]
	3GPP TS 38.508-1 [28]

The Test Suite in TTCN3 is contained in multiple ASCII files which accompany the present document.

A.2.2 E-UTRA LTE positioning Test Cases

Table A.2.2-1 lists all approved test cases.

Table A.2.2-1: E-UTRA LTE positioning TTCN test cases

Test case	Description
7.1.1	UE Network Capability
7.2.1.1	Location Notification
7.2.1.2	Privacy Verification – Location Allowed if no Response
7.2.1.3	Privacy Verification – Location not Allowed if No Response
7.2.2.2.5s	Basic Self Location: UE-assisted sub-test 5
7.2.2.2.6.FDDs	Basic Self Location: UE-assisted sub-test 6 (FDD)
7.2.2.2.6.TDDs	Basic Self Location: UE-assisted sub-test 6 (TDD)
7.3.1.1	Position Capability Transfer
7.3.2.1	LPP Duplicated Message
7.3.2.2	LPP Acknowledgment
7.3.2.3	LPP Retransmission
7.3.3.1	LPP Requested Method not Supported – UE-Assisted sub-test 1
7.3.4.2.5s	E-SMLC Initiated Assistance Data Delivery followed by Location Information Transfer: UE-Assisted sub-test 5
7.3.4.2.6s	E-SMLC Initiated Assistance Data Delivery followed by Location Information Transfer: UE-Assisted sub-test 6
7.3.4.2.7s	E-SMLC Initiated Assistance Data Delivery followed by Location Information Transfer: UE-Assisted sub-test 7
7.3.4.4.5s	E-SMLC Initiated Position Measurement without assistance data: UE-Assisted sub-test 5
7.3.4.4.7s	E-SMLC Initiated Position Measurement without assistance data: UE-Assisted sub-test 7
7.3.5.1.5s	E-SMLC initiated Abort sub-test 5

A.2.3 UTRA A-GNSS Test Cases

Table A.2.3-1 lists all approved test cases.

Table A.2.3-1: UTRA A-GNSS TTCN test cases

Test case	Description
6.2.1.1.4s	NI-LR Emergency Call: UE-Based A-GNSS sub-test 4
6.2.2.1.4s	MO-LR Position Estimate: UE-Based A-GNSS sub-test 4
6.2.2.3.4s	MO-LR Position Estimate: UE-Based A-GNSS – Failure Not Enough Satellites sub-test 4
6.2.3.2.4s	MT-LR Position Estimate: UE-Based A-GNSS – Failure Not Enough Satellites sub-test 4
6.2.3.3	Location Notification
6.2.3.4	Privacy Verification - Location Allowed if No Response
6.2.3.5	Privacy Verification - Location Not Allowed if No Response

A.2.4 NR/5GC Test Cases

Table A.2.4-1 lists all approved test cases.

Table A.2.4-1: NR/5GC TTCN test cases

Test case	Description
9.3.1.1.NR5GC	Position Capability Transfer
9.3.1.2.5s.NR5GC	LPP Abort: Subtest 5
9.3.1.2.15s.NR5GC	LPP Abort: Subtest 15
9.3.1.2.19s.NR5GC	LPP Abort: Subtest 19
9.3.1.2.20s.NR5GC	LPP Abort: Subtest 20
9.3.1.2.21s.NR5GC	LPP Abort: Subtest 21
9.3.2.1.NR5GC	LPP Duplicated Message
9.3.2.2.NR5GC	LPP Acknowledgement
9.3.2.3.NR5GC	LPP Retransmission
9.3.4.1.15s.NR5GC	E-SMLC Initiated Assistance Data Delivery followed by Location Information Transfer: UE-Based: Subtest 15
9.3.4.2.15s.NR5GC	E-SMLC Initiated Assistance Data Delivery followed by Location Information Transfer: UE-Assisted: Subtest 15
9.3.4.2.19s.NR5GC	E-SMLC Initiated Assistance Data Delivery followed by Location Information Transfer: UE-Assisted: Subtest 19
9.3.4.2.20s.NR5GC	E-SMLC Initiated Assistance Data Delivery followed by Location Information Transfer: UE-Assisted: Subtest 20
9.3.4.2.21s.NR5GC	E-SMLC Initiated Assistance Data Delivery followed by Location Information Transfer: UE-Assisted: Subtest 21
9.3.4.2.22s.NR5GC	E-SMLC Initiated Assistance Data Delivery followed by Location Information Transfer: UE-Assisted: Subtest 22
9.3.4.3.15s.NR5GC	E-SMLC Initiated Position Measurement without Assistance Data: UE-Based: Subtest 15
9.3.4.4.15s.NR5GC	E-SMLC Initiated Position Measurement without Assistance Data: UE-Assisted: Subtest 15
9.3.4.4.19s.NR5GC	E-SMLC Initiated Position Measurement without Assistance Data: UE-Assisted: Subtest 19
9.3.4.4.20s.NR5GC	E-SMLC Initiated Position Measurement without Assistance Data: UE-Assisted: Subtest 20
9.3.4.4.21s.NR5GC	E-SMLC Initiated Position Measurement without Assistance Data: UE-Assisted: Subtest 21

Annex B (informative): ASN.1 modules

B.1 A-GPS ASN.1 module

Refer to 3GPP TS 34.123-3 [8], annex H.1.

B.2 A-GNSS ASN.1 modules

Refer to 3GPP TS 34.123-3 [8], annex H.2.

B.3 LPP ASN.1

The LPP asn.1 modules are extracted from 3GPP TS 37.355 [10] without any change.

B.4 SLPP ASN.1

The SLPP asn.1 modules are extracted from 3GPP TS 38.355 [30] without any change.

Annex C (informative)
E-UTRA System Simulator Extension ASP Definitions in TTCN-3

C.1 EUTRA_POS_ASP_TypeDefs

Type definitions for configuration of Positioning Reference Signal in the system simulator;
Common design principles: refer to 3GPP TS 36.523-3 [6] annex D.

C.1.1 System_Configuration_PosExt

Formal ASP Definitions for EUTRA PRS system configuration

SystemRequest_PosExt_Type

TTCN-3 Union Type		
Name	**SystemRequest_PosExt_Type**	
Comment	NOTE: Same structure as SystemRequest_Type in 3GPP TS 36.523-3 [6] annex D	
Cell	CellConfigRequest_PosExt_Type	configure PRS in an LTE cell; the cell shall be configured as usual beforehand and SS shall raise an error when the cell does not exist yet; there is no explicit release of the PRS but the whole cell needs to be released

SystemConfirm_PosExt_Type

TTCN-3 Union Type		
Name	**SystemConfirm_PosExt_Type**	
Comment	confirmations for system configuration; in general to be sent after the configuration has been done	
Cell	Null_Type	(no further parameters from SS)

C.1.2 Cell_Configuration_PosExt

Specific Info for PRS Cell Configuration Primitive

C.1.2.1 Cell_Configuration_Common_PosExt

CellConfigRequest_PosExt_Type

TTCN-3 Union Type		
Name	**CellConfigRequest_PosExt_Type**	
Comment	NOTE: Same structure as SystemRequest_Type in 3GPP TS 36.523-3 [6] annex D	
AddOrReconfigure	CellConfigInfo_PosExt_Type	for PRS configuration of given cell: CellId : identifier of the cell to be configured RoutingInfo : None TimingInfo : Now (for initial configuration and for reconfiguration in general) ControlInfo : CnfFlag:=true; FollowOnFlag:=false (in general)

CellConfigInfo_PosExt_Type

TTCN-3 Record Type			
Name	**CellConfigInfo_PosExt_Type**		
Comment	NOTE: Same structure as SystemRequest_Type in 3GPP TS 36.523-3 [6] annex D common information for initial cell configuration or reconfiguration; in case of reconfiguration OMIT means 'keep configuration as it is'		
Basic	BasicCellConfig_PosExt_Type	opt	Extended PRS information for a cell

BasicCellConfig_PosExt_Type

TTCN-3 Record Type			
Name	**BasicCellConfig_PosExt_Type**		
Comment	NOTE: Same structure as SystemRequest_Type in 3GPP TS 36.523-3 [6] annex D Physical layer:		
PhysicalLayerConfigDL	PhysicalLayerConfigDL_PosExt_Type	opt	Default settings regarding physical control PRS signal

C.1.2.2 Downlink_Physical_Layer_Configuration_PosExt

Downlink physical layer configuration:
- DL antenna configuration for port 6.
- PRS configuration.

AntennaPort6_Config_Type

TTCN-3 Record Type			
Name	**AntennaPort6_Config_Type**		
Comment	to configure antenna port 6 for positioning; all fields are declared as optional to allow single reconfigurations; in this case omit means "keep as it is"		
Port	AntennaPort_Type	opt	
PRS	PRS_Info	opt	
RelativeTxPower	ToRS_EPRE_Ratio_Type	opt	power ratio for DI-PRS resource elements relative to the RS

PhysicalLayerConfigDL_PosExt_Type

TTCN-3 Record Type			
Name	**PhysicalLayerConfigDL_PosExt_Type**		
Comment	all fields are declared as optional to allow single reconfigurations; in this case omit means "keep as it is"		
AntennaPort6	AntennaPort6_Config_Type	opt	

C.1.3 System_Interface_PosExt

SYSTEM_CTRL_POSEXT_REQ

TTCN-3 Record Type			
Name	**SYSTEM_CTRL_POSEXT_REQ**		
Comment			
Common	ReqAspCommonPart_Type		For configuration of antenna port 6 TimingInfo is now
Request	SystemRequest_PosExt_Type		configure a PRS in a cell

SYSTEM_CTRL_POSEXT_CNF

TTCN-3 Record Type			
Name	**SYSTEM_CTRL_POSEXT_CNF**		
Comment			
Common	CnfAspCommonPart_Type		TimingInfo is ignored by TTCN => SS may set TimingInfo to "None"
Confirm	SystemConfirm_PosExt_Type		

EUTRA_SYSTEM_POSEXT_PORT

TTCN-3 Port Type		
Name	**EUTRA_SYSTEM_POSEXT_PORT**	
Comment	EUTRA PTC: Ext Port for system configuration of antenna port 6	
out	SYSTEM_CTRL_POSEXT_REQ	
in	SYSTEM_CTRL_POSEXT_CNF	

C.2 References to TTCN-3

References to TTCN-3		
EUTRA_POS_ASP_TypeDefs	EUTRA/EUTRA_POS_ASP_TypeDefs.ttcn	Rev 36622

Annex D (informative)
Positioning System Simulator ASP Definitions in TTCN-3

D.1 PosSystem_ASP_TypeDefs

PosSystem_ASP_TypeDefs: Constant Definitions

TTCN-3 Basic Types			
tsc_MaxPosSystems	integer	12	

PositioningSystemType

TTCN-3 Enumerated Type	
Name	**PositioningSystemType**
Comment	
gps	
modernizedGPS	
glonass	
galileo	
qzss	
otdoa	
ecid	
bds	
mbs	
wlan	
bluetooth	
sensor	

PositioningSystemList_Type

TTCN-3 Record of Type	
Name	**PositioningSystemList_Type**
Comment	
record length (1..tsc_MaxPosSystems) of PositioningSystemType	

Altitude_Type

TTCN-3 Record Type			
Name	**Altitude_Type**		
Comment			
Height	integer		Height above a sea level in meters

D.1.1 PosSystem_Load_Scenario

Load scenarios follow 3GPP TS 37.571-5 [5]

PosSystem_Load_Scenario: Basic Type Definitions

TTCN-3 Basic Types		
ScenarioNumber_Type	integer	

PositioningScenario_Type

TTCN-3 Union Type		
Name	**PositioningScenario_Type**	
Comment		
LTE_Positioning	ScenarioNumber_Type	Reference to 3GPP TS 37.571-5 [5]
AGNSS	ScenarioNumber_Type	Reference to 3GPP TS 37.571-5 [5]
V2X	ScenarioNumber_Type	V2X Reference to 3GPP TS 36.508 [7] clause 4.11
NR_Positioning	ScenarioNumber_Type	Reference to 3GPP TS 37.571-5 [5]
Aerial	ScenarioNumber_Type	Reference to 3GPP TS 36.508 [7] clause 4.12
NR_Sidelink	ScenarioNumber_Type	Reference to 3GPP TS 38.508-1 [30] clause 4.11

D.1.2 PosSystem_Retrieve_Data

D.1.2.1 SatSystem_Retrieve_UTRAN_Data

UTRAN_GPS_AssistanceDataRequest_Type

TTCN-3 Record Type			
Name	**UTRAN_GPS_AssistanceDataRequest_Type**		
Comment			
AssistanceDataReqGPS	bitstring		bitstring containing 3GPP TS 25.331 [12] type UE-Positioning-GPS-AdditionalAssistanceDataRequest

UTRAN_GANSS_AssistanceDataRequest_Type

TTCN-3 Record Type			
Name	**UTRAN_GANSS_AssistanceDataRequest_Type**		
Comment			
AssistanceDataReqGANSS	bitstring	opt	bitstring containing 3GPP TS 25.331 [12] type UE-Positioning-GANSS-AdditionalAssistanceDataRequest
AssistanceDataReqGANSSv860ext	bitstring	opt	bitstring containing 3GPP TS 25.331 [12] type UE-Positioning-GANSS-AdditionalAssistanceDataRequest-v860ext
AssistanceDataReqGANSSvc50ext	bitstring	opt	bitstring containing 3GPP TS 25.331 [12] type UE-Positioning-GANSS-AdditionalAssistanceDataRequest-vc50ext

UTRAN_AssistanceDataRequest_Type

TTCN-3 Union Type		
Name	**UTRAN_AssistanceDataRequest_Type**	
Comment		
GPS	UTRAN_GPS_AssistanceDataRequest_Type	
GANSS	UTRAN_GANSS_AssistanceDataRequest_Type	

UTRAN_AssistanceDataRequestList_Type

TTCN-3 Record of Type	
Name	**UTRAN_AssistanceDataRequestList_Type**
Comment	
record length(1..tsc_MaxPosSystems) of UTRAN_AssistanceDataRequest_Type	

UTRAN_GPS_AssistanceData_Type

TTCN-3 Record Type			
Name	**UTRAN_GPS_AssistanceData_Type**		
Comment	The returned Almanac information is split into two fields: Almanac for satellites 1 to 16 in AssistanceDataGPSr7 together with other information; Almanac for satellites 17 to 31: in 'almanacSat17To31'		
AssistanceDataGPSr7	bitstring	opt	bitstring containing 3GPP TS 25.331 [12] type UE-Positioning-GPS-AssistanceData-r7
AssistanceDataGPSr12	bitstring	opt	bitstring containing 3GPP TS 25.331 [12] type UE-Positioning-GPS-AssistanceData-r12
AlmanacSatInfoList17To31	bitstring	opt	bitstring containing 3GPP TS 25.331 [12] type AlmanacSatInfoList

UTRAN_GANSS_AssistanceData_Type

TTCN-3 Record Type			
Name	**UTRAN_GANSS_AssistanceData_Type**		
Comment			
AssistanceDataGANSS_r8	bitstring	opt	bitstring containing 3GPP TS 25.331 [12] type UE-Positioning-GANSS-AssistanceData-r8
AssistanceDataGANSS_r12	bitstring	opt	bitstring containing 3GPP TS 25.331 [12] type UE-Positioning-GANSS-AssistanceData-r12

UTRAN_AssistanceData_Type

TTCN-3 Union Type		
Name	**UTRAN_AssistanceData_Type**	
Comment		
GPS	UTRAN_GPS_AssistanceData_Type	
GANSS	UTRAN_GANSS_AssistanceData_Type	

UTRAN_AssistanceDataList_Type

TTCN-3 Record of Type	
Name	**UTRAN_AssistanceDataList_Type**
Comment	
record length(1..tsc_MaxPosSystems) of UTRAN_AssistanceData_Type	

D.1.2.2 PosSystem_Retrieve_LPP_Data

When data is to be requested for more than one positioning system, they will be included into one single template. This template will include a list with one element for each positioning system.
See template "cs_RequestAssistData_UEB_GpsAndGlonass" as an example.

LPP_AssistanceDataRequest_Type

TTCN-3 Union Type		
Name	**LPP_AssistanceDataRequest_Type**	
Comment		
LPP_AssistanceData	bitstring	bitstring containing 3GPP TS 37.355 [10] type RequestAssistanceData-r9-IEs

LPP_AssistanceDataProvide_Type

TTCN-3 Union Type		
Name	**LPP_AssistanceDataProvide_Type**	
Comment		
LPP_AssistanceData	bitstring	bitstring containing 3GPP TS 37.355 [10] type ProvideAssistanceData-r9-IEs

LPP_AssistanceDataRequestList_Type

TTCN-3 Record of Type	
Name	**LPP_AssistanceDataRequestList_Type**
Comment	
record length(1..tsc_MaxPosSystems) of LPP_AssistanceDataRequest_Type	

LPP_AssistanceDataList_Type

TTCN-3 Record of Type	
Name	**LPP_AssistanceDataList_Type**
Comment	
record length(1..tsc_MaxPosSystems) of LPP_AssistanceDataProvide_Type	

D.1.2.3 PosSystem_Retrieve_Data_Common

AssistanceDataRetrieveRequest_Type

TTCN-3 Union Type		
Name	**AssistanceDataRetrieveRequest_Type**	
Comment		
LPP	LPP_AssistanceDataRequestList_Type	LTE/NR positioning
UTRA_GNSS	UTRAN_AssistanceDataRequestList_Type	Retrieve A-GNSS data list in the order of positioning system provided in PositioningSystemList

AssistanceDataRetrieveResponse_Type

TTCN-3 Union Type		
Name	**AssistanceDataRetrieveResponse_Type**	
Comment		
LPP	LPP_AssistanceDataList_Type	LTE/NR positioning
UTRA_AGNSS	UTRAN_AssistanceDataList_Type	Retrieve A-GNSS data list in the order of positioning system provided in PositioningSystemList

D.1.3 PosSystem_PowerControl

PowerControl_Type

TTCN-3 Union Type		
Name	**PowerControl_Type**	
Comment	V2X. The GNSS simulator is switched on/off without stopping the UTC time sequence	
PowerOn	Null_Type	By default the power is on in the GNSS simulator
PowerOff	Null_Type	

D.1.4 PosSystem_GnssUtcTime

GnssUtcTime_Type

TTCN-3 Record Type			
Name	**GnssUtcTime_Type**		
Comment			
Struct_tm	Struct_tm_Type		Local GNSS time
TimezoneInfo	integer		Difference (in seconds) between the UTC time (GMT) and the local time

D.1.5 System_Interface

Common ASP definitions

PosSystemCrtlRequest_Type

TTCN-3 Union Type		
Name	**PosSystemCrtlRequest_Type**	
Comment		
Start	Null_Type	
Stop	Null_Type	
LoadScenario	PositioningScenario_Type	
RetrieveData	AssistanceDataRetrieveRequest_Type	
TriggerMove	Null_Type	V2X. Start moving to the next area/zone in the scenario
TriggerPowerOnOff	PowerControl_Type	V2X. Switch on/off the transmitter without stopping UTC time sequence
SetAltitude	Altitude_Type	Aerial Set the height above the sea level to the provided value
TriggerAerialMove	Altitude_Type	Aerial Start moving towards target height
RetrieveGnssUtcTime	Null_Type	Get current UTC time of the GNSS scenario. This can only be used once a GNSS scenario has been loaded and started

PosSystemCrtlConfirm_Type

TTCN-3 Union Type		
Name	**PosSystemCrtlConfirm_Type**	
Comment	In general, to be sent after the configuration has been done	
Start	Null_Type	
Stop	Null_Type	
LoadScenario	Null_Type	
RetrieveData	AssistanceDataRetrieveResponse_Type	
TriggerMove	Null_Type	V2X
TriggerPowerOnOff	Null_Type	V2X
SetAltitude	Null_Type	Aerial
TriggerAerialMove	Null_Type	Aerial
RetrieveGnssUtcTime	GnssUtcTime_Type	

POS_SYSTEM_CTRL_REQ

TTCN-3 Record Type			
Name	**POS_SYSTEM_CTRL_REQ**		
Comment			
PositioningSystemList	PositioningSystemList_Type		The list of positioning systems to be configured; there is only one system of one type in the list e.g. gps, galileo etc.
CnfFlag	boolean		true => SS shall send CNF
Request	PosSystemCrtlRequest_Type		

POS_SYSTEM_CTRL_CNF

TTCN-3 Record Type			
Name	**POS_SYSTEM_CTRL_CNF**		
Comment			
PositioningSystemList	PositioningSystemList_Type		
Confirm	PosSystemCrtlConfirm_Type		

POS_MTC_SYSTEM_PORT

TTCN-3 Port Type		
Name	**POS_MTC_SYSTEM_PORT**	
Comment	Positioning: port at the MTC to be mapped to the system	
out	POS_SYSTEM_CTRL_REQ	
in	POS_SYSTEM_CTRL_CNF	

POS_MTC_PTC_PORT

TTCN-3 Port Type		
Name	**POS_MTC_PTC_PORT**	
Comment	Positioning: port at the MTC to hand over information from the positioning system to the PTC(s)	
out	POS_SYSTEM_CTRL_CNF	
in	POS_SYSTEM_CTRL_REQ	

POS_PTC_MTC_PORT

TTCN-3 Port Type		
Name	**POS_PTC_MTC_PORT**	
Comment	Positioning: port at the PTC(s) to send commands to the positioning system via the MTC	
out	POS_SYSTEM_CTRL_REQ	
in	POS_SYSTEM_CTRL_CNF	

D.2 CommonDefs

CommonDefs: Basic Type Definitions

TTCN-3 Basic Types		
Null_Type	boolean (true)	dummy type for 'typeless' fields in unions

Struct_tm_Type

TTCN-3 Record Type			
Name	**Struct_tm_Type**		
Comment	TTCN-3 equivalent for the "struct tm" as defined on C standard library (time.h or ctime.h for C or C++)		
tm_sec	integer		seconds after the minute 0..61 (tm_sec is generally 0-59. Extra range to accommodate for leap seconds in certain systems)
tm_min	integer		minutes after the hour 0..59
tm_hour	integer		hours since midnight 0..23
tm_mday	integer		day of the month 1..31
tm_mon	integer		months since January 0..11
tm_year	integer		years since 1900
tm_wday	integer		days since Sunday 0..6
tm_yday	integer		days since January 1 0..365
tm_isdst	integer		Daylight Saving Time flag

D.3 References to TTCN-3

References to TTCN-3		
PosSystem_ASP_TypeDefs	POS/PosSystem_ASP_TypeDefs.ttcn	Rev 32599
CommonDefs	Common/CommonDefs.ttcn	Rev 32608

Annex E (informative)
NR5GC System Simulator Extension ASP Definitions in TTCN-3

E.1 NR5GC_POS_ASP_TypeDefs

Type definitions for configuration of DL Positioning Reference Signal in the system simulator;
Common design principles: refer to 3GPP TS 38.523-3 [24] annex D.

E.1.1 NR_System_Configuration_PosExt

Formal ASP Definitions for NR5GC DL-PRS system configuration

NR_SystemRequest_PosExt_Type

TTCN-3 Union Type		
Name	**NR_SystemRequest_PosExt_Type**	
Comment	NOTE: Same structure as NR_SystemRequest_Type in 3GPP TS 38.523-3 [24] annex D	
Cell	NR_CellConfigRequest_PosExt_Type	configure DL-PRS in an NR cell; the cell shall be configured as usual beforehand and SS shall raise an error when the cell does not exist yet; there is no explicit release of the PRS but the whole cell needs to be released

NR_SystemConfirm_PosExt_Type

TTCN-3 Union Type		
Name	**NR_SystemConfirm_PosExt_Type**	
Comment	confirmations for system configuration; in general to be sent after the configuration has been done	
Cell	Null_Type	(no further parameters from SS)

E.1.2 NR_Cell_Configuration_PosExt

Specific Info for DL-PRS Cell Configuration Primitive

E.1.2.1 NR_Cell_Configuration_Common_PosExt

NR_CellConfigRequest_PosExt_Type

TTCN-3 Union Type		
Name	**NR_CellConfigRequest_PosExt_Type**	
Comment	NOTE: Same structure as NR_SystemRequest_Type in 3GPP TS 38.523-3 [24] annex D	
AddOrReconfigure	NR_CellConfigInfo_PosExt_Type	for PRS configuration of given cell: CellId : identifier of the cell to be configured RoutingInfo : None RlcBearerRouting : 'None' MacBearerRouting : 'omit' TimingInfo : Now (for initial configuration and for reconfiguration in general) ControlInfo : CnfFlag:=true; FollowOnFlag:=false (in general)

NR_CellConfigInfo_PosExt_Type

TTCN-3 Record Type			
Name	**NR_CellConfigInfo_PosExt_Type**		
Comment	NOTE: Same structure as NR_SystemRequest_Type in 3GPP TS 38.523-3 [24] annex D common information for initial cell configuration or reconfiguration; in case of reconfiguration OMIT means 'keep configuration as it is'		
PhysicalLayer	NR_CellConfigPhysicalLayer_PosExt_Type	opt	Extended DL-PRS information for a cell

NR_CellConfigPhysicalLayer_PosExt_Type

TTCN-3 Record Type			
Name	**NR_CellConfigPhysicalLayer_PosExt_Type**		
Comment	NOTE: Same structure as NR_SystemRequest_Type in 3GPP TS 38.523-3 [24] annex D Physical layer:		
Downlink	NR_CellConfigPhysicalLayerDownlink_PosExt_Type	opt	Default settings regarding physical control DL-PRS signal

E.1.2.2 Downlink_Physical_Layer_Configuration_PosExt

Downlink physical layer configuration:
- DL-PRS Frequency Layer configuration.
- DL-PRS TRP configuration.

Downlink_Physical_Layer_Configuration_PosExt: Basic Type Definitions

TTCN-3 Basic Types		
DlPrs_CombSizeN_Type	NR_DL_PRS_PositioningFrequencyLayer_r16.dl_PRS_CombSizeN_r16	

DlPrs_FreqLayerConfig_Type

TTCN-3 Record Type			
Name	**DlPrs_FreqLayerConfig_Type**		
Comment			
DlPrs_ResourceBW	integer	opt	
DlPrs_CombSizeN	DlPrs_CombSizeN_Type	opt	

DlPrs_TRPConfig_Type

TTCN-3 Record Type			
Name	**DlPrs_TRPConfig_Type**		
Comment			
NrDlPrsSfn0Offset	NR_DL_PRS_SFN0_Offset_r16	opt	
NrDlPrsExpectedRstd	integer	opt	
NrDlPrsExpectedRstdUncertainty	integer	opt	
NrDlPrsInfo	NR_DL_PRS_Info_r16	opt	

DlPrs_EPREs_Type

TTCN-3 Record Type			
Name	**DlPrs_EPREs_Type**		
Comment	EPRE for DL-PRS relative to SSS		
DlPrsToSSS	NR_EPRE_Ratio_Type	opt	Positioning reference signal; 38.211 clause 7.4.1.7

DL_PRSConfig_Type

TTCN-3 Record Type			
Name	**DL_PRSConfig_Type**		
Comment	to configure DL-PRS for positioning; all fields are declared as optional to allow single reconfigurations; in this case omit means "keep as it is"		
DlPrs_FreqLayerConfig	DlPrs_FreqLayerConfig_Type	opt	
DlPrs_TRPConfig	DlPrs_TRPConfig_Type	opt	
RelativeTxPower	DlPrs_EPREs_Type	opt	transmit power for DL-PRS signal

NR_CellConfigPhysicalLayerDownlink_PosExt_Type

TTCN-3 Record Type			
Name	**NR_CellConfigPhysicalLayerDownlink_PosExt_Type**		
Comment	all fields are declared as optional to allow single reconfigurations; in this case omit means "keep as it is"		
DL_PRSConfig	DL_PRSConfig_Type	opt	

E.1.3 NR_System_Interface_PosExt

NR_SYSTEM_CTRL_POSEXT_REQ

TTCN-3 Record Type			
Name	**NR_SYSTEM_CTRL_POSEXT_REQ**		
Comment			
Common	NR_ReqAspCommonPart_Type		For configuration of DL-PRS TimingInfo is now
Request	NR_SystemRequest_PosExt_Type		configure a DL-PRS in a cell

NR_SYSTEM_CTRL_POSEXT_CNF

TTCN-3 Record Type			
Name	**NR_SYSTEM_CTRL_POSEXT_CNF**		
Comment			
Common	NR_CnfAspCommonPart_Type		TimingInfo is ignored by TTCN => SS may set TimingInfo to "None"
Confirm	NR_SystemConfirm_PosExt_Type		

NR_SYSTEM_POSEXT_PORT

TTCN-3 Port Type		
Name	**NR_SYSTEM_POSEXT_PORT**	
Comment	NR PTC: Ext Port for system configuration of DL-PRS	
out	NR_SYSTEM_CTRL_POSEXT_REQ	
in	NR_SYSTEM_CTRL_POSEXT_CNF	

E.2 References to TTCN-3

References to TTCN-3		
NR5GC_POS_ASP_TypeDefs	NR5GC/NR5GC_POS_ASP_TypeDefs.ttcn	Rev 36621

Annex F (informative)
NR Sidelink System Simulator Extension ASP Definitions in TTCN-3

F.1 NR5GC_SL_POS_ASP_TypeDefs

Type definitions for configuration of SL Positioning Reference Signal in the system simulator;
Common design principles: refer to 3GPP TS 38.523-3 [24] annex D.

F.1.1 NR_SL_System_Configuration_PosExt

Formal ASP Definitions for NR5GC SL-PRS system configuration

NR_SL_SystemRequest_PosExt_Type

TTCN-3 Union Type		
Name	**NR_SL_SystemRequest_PosExt_Type**	
Comment	NOTE: Same structure as NR_SL_SystemRequest_Type in 3GPP TS 38.523-3 [24] annex D	
SS_UE_Config	NR_SL_ConfigRequest_PosExt_Type	configure SL-PRS in a NR-SS-UE; the NR-SS-UE shall be configured as usual beforehand and SS shall raise an error when the NR-SS-UE does not exist yet; there is no explicit release of the SL-PRS but the whole NR-SS-UE needs to be released

NR_SL_SystemConfirm_PosExt_Type

TTCN-3 Union Type		
Name	**NR_SL_SystemConfirm_PosExt_Type**	
Comment	confirmations for system configuration; in general to be sent after the configuration has been done	
SS_UE_Config	Null_Type	(no further parameters from SS)

F.1.2 NR_SL_SS_UE_Config_PosExt

Specific Info for SL-PRS NR-SS-UE Configuration Primitive

NR_SL_ConfigRequest_PosExt_Type

TTCN-3 Union Type		
Name	**NR_SL_ConfigRequest_PosExt_Type**	
Comment	NOTE: Same structure as NR_SL_SystemRequest_Type in 3GPP TS 38.523-3 [24] annex D	
AddOrReconfigure	NR_SL_ConfigInfo_PosExt_Type	For SL-PRS configuration of given NR-SS-UE: SS_UE_Id : identifier of the NR-SS-UE to be configured RoutingInfo : None TimingInfo : 'Now'(for initial configuration and for reconfiguration in general) ControlInfo : FollowOnFlag:=false

NR_SL_ConfigInfo_PosExt_Type

TTCN-3 Record Type			
Name	**NR_SL_ConfigInfo_PosExt_Type**		
Comment			
SL_BWP_PoolConfigList	NR_SL_BWP_PoolConfigList_PosExt_Type	opt	Extended SL-PRS information for a NR-SS-UE

NR_SL_BWP_PoolConfigList_PosExt_Type

TTCN-3 Record of Type	
Name	**NR_SL_BWP_PoolConfigList_PosExt_Type**
Comment	
record of NR_SL_BWP_PoolConfig_PosExt_Type	

NR_SL_BWP_PoolConfig_PosExt_Type

TTCN-3 Record Type			
Name	**NR_SL_BWP_PoolConfig_PosExt_Type**		
Comment			
SL_PRS_Config	NR_SL_PRS_Config_Type		

NR_SL_PRS_Config_Type

TTCN-3 Record Type			
Name	**NR_SL_PRS_Config_Type**		
Comment	to configure SL-PRS for positioning; all fields are declared as optional to allow single reconfigurations; in this case omit means "keep as it is"		
SL_PRS_Pool	NR_SL_PRS_Pool_Type	opt	configuration of the resource pool for SL-PRS
SL_PRS_Info	NR_SL_PRS_Info_Type	opt	parameters needed for SL-PRS sequence generation
RelativeTxPower	NR_SL_PRS_EPREs_Type	opt	transmit power for the SL-PRS signal

NR_SL_PRS_Info_Type

TTCN-3 Record Type			
Name	**NR_SL_PRS_Info_Type**		
Comment			
SL_PRS_SequenceID	integer	opt	Value as provided in SL-PRS-AssistanceData; i.e. 0 for NR-SS-UE1, 1 for NR-SS-UE2 and 2 for NR-SS-UE3

NR_SL_PRS_EPREs_Type

TTCN-3 Record Type			
Name	**NR_SL_PRS_EPREs_Type**		
Comment	EPRE for SL-PRS relative to NR-SS-UE power		
SlPrsToNrSsUe	NR_EPRE_Ratio_Type	opt	Sidelnk Positioning Reference Signal; TS 38.211 clause 8.4.1.6

F.1.2.1 NR_SL_PRS_Pool_Configuration

NR_SL_PRS_Pool_Configuration: Basic Type Definitions

TTCN-3 Basic Types		
NR_SL_PRS_NumberOfSymbols_Type	SL_PRS_ResourceSharedSL_PRS_RP_r18.mNumberOfSymbols_r18	
NR_SL_PRS_CombSizeN_AndReOffset_Type	SL_PRS_ResourceSharedSL_PRS_RP_r18.sl_PRS_CombSizeN_AndReOffset_r18	
NR_SL_PRS_NumSymSCI2D_Type	SL_ResourcePool_r16.numSym_SL_PRS_2ndStageSCI_r18	
NR_SL_PRS_PSCCH_Config_Type	SL_PRS_ResourcePool_r18.sl_PRS_PSCCH_Config_r18	
NR_SL_PRS_StartRB_Subchannel_Type	SL_PRS_ResourcePool_r18.sl_StartRB_SubchannelDedicatedSL_PRS_RP_r18	
NR_SL_PRS_RB_Number_Type	SL_PRS_ResourcePool_r18.sl_RB_Number_r18	
NR_SL_TimeResource_r18_Type	SL_PRS_ResourcePool_r18.sl_TimeResource_r18	
NR_SL_PRS_ResourceReservePeriodList_Type	SL_PRS_ResourcePool_r18.sl_PRS_ResourceReservePeriodList_r18	
NR_SL_PRS_NumSubchannel_Type	SL_PRS_ResourcePool_r18.sl_NumSubchannelDedicatedSL_PRS_RP_r18	
NR_SL_PRS_SubchannelSize_Type	SL_PRS_ResourcePool_r18.sl_SubchannelSizeDedicatedSL_PRS_RP_r18	
NR_SL_PRS_NumBitsSCI1B_Type	SL_PRS_ResourcePool_r18.sl_NumReservedBitsSCI1B_DedicatedSL_PRS_RP_r18	

NR_SL_PRS_Pool_Type

TTCN-3 Union Type		
Name	**NR_SL_PRS_Pool_Type**	
Comment		
SharedPool	NR_SL_PRS_SharedPool_Type	
DedicatedPool	NR_SL_PRS_DedicatedPool_Type	

NR_SL_PRS_SharedPool_Type

TTCN-3 Record Type			
Name	**NR_SL_PRS_SharedPool_Type**		
Comment			
SL_PRS_SharedResourcePool	NR_SL_PRS_SharedResourcePool_Type		
SL_PRS_SharedSCI	NR_SL_PRS_SharedSCI_Type		

NR_SL_PRS_DedicatedPool_Type

TTCN-3 Record Type			
Name	**NR_SL_PRS_DedicatedPool_Type**		
Comment			
SL_PRS_DedicatedResourcePool	NR_SL_PRS_DedicatedResourcePool_Type		
SL_PRS_DedicatedSCI	NR_SL_PRS_DedicatedSCI_Type		

NR_SL_PRS_SharedResourcePool_Type

TTCN-3 Record Type			
Name	**NR_SL_PRS_SharedResourcePool_Type**		
Comment	these types add up to those already configured for the SL communication resource pool with SL_TxPool.SL_ResourcePool		
SL_TimeResource	NR_SL_TimeResource_Type	opt	
SL_PRS_NumberOfSymbols	NR_SL_PRS_NumberOfSymbols_Type	opt	
SL_PRS_CombSizeN_AndReOffset	NR_SL_PRS_CombSizeN_AndReOffset_Type	opt	
SL_PRS_NumSymSCI2D	NR_SL_PRS_NumSymSCI2D_Type	opt	

NR_SL_PRS_SharedSCI_Type

TTCN-3 Record Type			
Name	**NR_SL_PRS_SharedSCI_Type**		
Comment			
PSCCH_Sci	NR_SL_PRS_PSCCH_SharedSCI	opt	1st-stage SCI format for SL-PRS in Shared Resource Pool
PSSCH_Sci	NR_SL_PRS_PSSCH_SharedSCI	opt	2nd-stage SCI format for SL-PRS in Shared Resource Pool

NR_SL_PRS_PSCCH_SharedSCI

TTCN-3 Union Type		
Name	**NR_SL_PRS_PSCCH_SharedSCI**	
Comment		
SciFormat1A	NR_SL_SciFormat1A_Type	Acc to TS 38.212 cl 8.3.1.1 SCI format 1-A

NR_SL_PRS_PSSCH_SharedSCI

TTCN-3 Union Type		
Name	**NR_SL_PRS_PSSCH_SharedSCI**	
Comment		
SciFormat2D	NR_SL_SciFormat2D_Type	Acc to TS 38.212 cl 8.4.1.4 SCI format 2-D

NR_SL_PRS_ResourceID_Type

TTCN-3 Union Type		
Name	**NR_SL_PRS_ResourceID_Type**	
Comment	TS 38.212 clause 8.4.1.4: SL PRS resource ID	
None	Null_Type	0 bit if only one SL-PRS-ResourceID configured by sl-PRS-ResourcesSharedSL-PRS-RP
Value	integer	number of bits depending on the number of entries in the higher layer parameter sl-PRS-ResourcesSharedSL-PRS-RP

NR_SL_PRS_RequestSci2D_Type

TTCN-3 Union Type		
Name	**NR_SL_PRS_RequestSci2D_Type**	
Comment	TS 38.212 clause 8.4.1.4: SL PRS request	
None	Null_Type	0 bit if the higher layer parameter in sl-SCI-basedSL-PRS-TxTriggerSCI2-D is not provided
Value	B1_Type	1 bit if the higher layer parameter in sl-SCI-basedSL-PRS-TxTriggerSCI2-D is provided

NR_SL_SciFormat2D_Type

TTCN-3 Record Type			
Name	**NR_SL_SciFormat2D_Type**		
Comment	Acc to TS 38.212 cl 8.4.1.4 SCI format 2-D For all fields: 'omit' means that the information shall not be contained in the SCI		
SL_PRS_ResourceID	NR_SL_PRS_ResourceID_Type	opt	
SL_PRS_Request	NR_SL_PRS_RequestSci2D_Type	opt	
EmbeddedSCI_Format	B2_Type	opt	According to TS 38.212, Table 8.4.1.4-1
EmbeddedSCI_Payload	NR_SL_PRS_Sci2D_EmbeddedSCI_Payload_Type	opt	number of bits determined according to TS 38.212, Table 8.4.1.4-1

NR_SL_PRS_Sci2D_EmbeddedSCI_Payload_Type

TTCN-3 Union Type		
Name	**NR_SL_PRS_Sci2D_EmbeddedSCI_Payload_Type**	
Comment		
SciFormat2A_Payload	NR_SL_SciFormat2A_Type	Acc to TS 38.212 cl 8.4.1.1 SCI format 2-A
SciFormat2B_Payload	NR_SL_SciFormat2B_Type	Acc to TS 38.212 cl 8.4.1.2 SCI format 2-B

NR_SL_PRS_DedicatedResourcePool_Type

TTCN-3 Record Type			
Name	**NR_SL_PRS_DedicatedResourcePool_Type**		
Comment			
SL_PRS_PSCCH_Config	NR_SL_PRS_PSCCH_Config_Type	opt	
SL_PRS_StartRB_Subchannel	NR_SL_PRS_StartRB_Subchannel_Type	opt	
SL_RB_Number	NR_SL_PRS_RB_Number_Type	opt	
SL_TimeResource	NR_SL_TimeResource_r18_Type	opt	
SL_PRS_ResourceReservePeriodList	NR_SL_PRS_ResourceReservePeriodList_Type	opt	
SL_PRS_Resource	SL_PRS_ResourceDedicatedSL_PRS_RP_r18	opt	
SL_PRS_NumSubchannel	NR_SL_PRS_NumSubchannel_Type	opt	
SL_PRS_SubchannelSize	NR_SL_PRS_SubchannelSize_Type	opt	
SL_PRS_NumBitsSCI1B	NR_SL_PRS_NumBitsSCI1B_Type	opt	

NR_SL_PRS_DedicatedSCI_Type

TTCN-3 Record Type			
Name	**NR_SL_PRS_DedicatedSCI_Type**		
Comment			
PSCCH_Sci	NR_SL_PRS_PSCCH_DedicatedSCI	opt	1st-stage SCI format for SL-PRS in Dedicated Resource Pool

NR_SL_PRS_PSCCH_DedicatedSCI

TTCN-3 Union Type		
Name	**NR_SL_PRS_PSCCH_DedicatedSCI**	
Comment		
SciFormat1B	NR_SL_SciFormat1B_Type	Acc to TS 38.212 cl 8.3.1.2 SCI format 1-B

NR_SL_SciFormat1B_Type

TTCN-3 Record Type			
Name	**NR_SL_SciFormat1B_Type**		
Comment	According to TS 38.212 clause 8.3.1.2 SCI format 1-B Source ID and Destination ID provided in NR_SL_MAC_LogicalChannelConfig_Type		
Priority	B3_Type	opt	Priority - 3 bits as specified in clause 5.7 of TS 23.586 and clause 5.22 of TS 38.321. Value '000' of Priority field corresponds to priority value '1', value '001' of Priority field corresponds to priority value '2', and so on.
CastType	B2_Type	opt	Cast type indicator â€“ 2 bits as defined in Table 8.3.1.2-1 and in clause 16.4A of TS 38.213
ResourceReservationPeriod	NR_SL_PRS_ResourceReservationPeriod_Type	opt	
TimeResourceAssignment	integer	opt	
ResourceIdIndication	NR_SL_PRS_ResourceIdIndication_Type	opt	
SL_PRS_Request	NR_SL_PRS_RequestSci1B_Type	opt	

NR_SL_PRS_ResourceReservationPeriod_Type

TTCN-3 Union Type		
Name	**NR_SL_PRS_ResourceReservationPeriod_Type**	
Comment	TS 38.212 clause 8.3.1.2: Resource reservation period	
None	Null_Type	0 bit if higher layer parameter sl-PRS-ResourceReservePeriodList is not configured
Value	integer	number of bits depending on the number of entries in the higher layer parameter sl-PRS-ResourceReservePeriodList, if configured

NR_SL_PRS_ResourceIdIndication_Type

TTCN-3 Union Type		
Name	**NR_SL_PRS_ResourceIdIndication_Type**	
Comment	TS 38.212 clause 8.3.1.2: Resource ID indication	
None	Null_Type	0 bit when only one SL PRS resource is configured within a slot in a dedicated SL PRS resource pool
Value	integer	number of bits depending on the value of higher layer parameter sl-MaxNumPerReserveDedicatedSL-PRS-RP

NR_SL_PRS_RequestSci1B_Type

TTCN-3 Union Type		
Name	**NR_SL_PRS_RequestSci1B_Type**	
Comment	TS 38.212 clause 8.3.1.2: SL PRS Request	
None	Null_Type	0 bit if higher layer parameter sl-SCI-basedSL-PRS-TxTriggerSCI1-B is not provided
Value	B1_Type	According to clause 8.4.4 of TS 38.214 when the higher layer parameter sl-SCI-basedSL-PRS-TxTriggerSCI1-B is provided

F.1.3 NR_Sidelink_System_Interface_PosExt

NR_SL_SYSTEM_CTRL_POSEXT_REQ

TTCN-3 Record Type			
Name	**NR_SL_SYSTEM_CTRL_POSEXT_REQ**		
Comment			
Common	NR_SL_ReqAspCommonPart_Type		For configuration of SL-PRS TimingInfo is now
Request	NR_SL_SystemRequest_PosExt_Type		configure a SL-PRS in a NR-SS-UE

NR_SL_SYSTEM_CTRL_POSEXT_CNF

TTCN-3 Record Type			
Name	**NR_SL_SYSTEM_CTRL_POSEXT_CNF**		
Comment			
Common	NR_SL_CnfAspCommonPart_Type		TimingInfo is ignored by TTCN => SS may set TimingInfo to "None"
Confirm	NR_SL_SystemConfirm_PosExt_Type		

NR_SL_SYSTEM_POSEXT_PORT

TTCN-3 Port Type		
Name	**NR_SL_SYSTEM_POSEXT_PORT**	
Comment	SL PTC: Ext Port for system configuration of SL-PRS	
out	NR_SL_SYSTEM_CTRL_POSEXT_REQ	
in	NR_SL_SYSTEM_CTRL_POSEXT_CNF	

F.2 References to TTCN-3

References to TTCN-3		
NR5GC_SL_POS_ASP_TypeDefs	NR5GC/NR5GC_SL_POS_ASP_TypeDefs.ttcn	Rev 36621

Annex G (informative): Change history

Change history

Date	TSG #	TSG Doc.	CR	Rev	Subject/Comment	Old	New
2011-06	RAN5#51	R5-112203	-	-	Initial proposal for A-GNSS		0.0.1
2011-09	RAN5#52	R5-113037	-	-	Introduced positioning test model	0.0.1	0.1.0
2011-11	RAN5#53	R5-115106	-	-	Imported A-GPS test model from 3GPP TS 34.123-3	0.1.0	1.0.0
2011-12	RAN#54	RP-111627	-	-	Presentation of V1.0.0	0.1.0	1.0.0
2012-05	RAN5#55	R5-121083	-	-	Presentation of V1.1.0	1.0.0	1.1.0
2012-06	RAN#56	RP-120614	-	-	Presentation of V2.0.0	1.0.0	2.0.0
2012-06	RAN#56	-	-	-	Editorial corrections and formattings.	2.0.0	9.0.0
2012-06	RAN#56	-	-	-	Raised to v10.0.0 with no change.	9.0.0	10.0.0
2012-09	RAN#57	R5-123690	0001	-	37571-4: Routine maintenance and updates	10.0.0	10.1.0
2013-06	RAN#60	R5-131136	0003	-	37571-4: Routine maintenance and updates	10.1.0	10.2.0
2013-06	RAN#60	R5s130125	0004	-	Addition of EUTRA UE Positioning test case 7.1.1	10.1.0	10.2.0
2013-06	RAN#60	R5s130145	0005	-	Addition of EUTRA UE Positioning test case 7.3.1.1	10.1.0	10.2.0
2013-06	RAN#60	R5s130147	0006	-	Addition of EUTRA UE Positioning test case 7.3.2.1	10.1.0	10.2.0
2013-06	RAN#60	R5s130149	0007	-	Addition of EUTRA UE Positioning test case 7.3.2.2	10.1.0	10.2.0
2013-06	RAN#60	R5s130151	0008	-	Addition of EUTRA UE Positioning test case 7.3.2.3	10.1.0	10.2.0
2013-06	RAN#60	R5s130153	0009	-	Addition of EUTRA UE Positioning test case 7.3.5.1_6s	10.1.0	10.2.0
2013-06	RAN#60	R5s130155	0010	-	Addition of EUTRA UE Positioning test case 7.3.5.1_5s	10.1.0	10.2.0
2013-06	RAN#60	R5s130157	0011	-	Addition of EUTRA UE Positioning test case 7.3.4.2_5s	10.1.0	10.2.0
2013-06	RAN#60	R5s130159	0012	-	Addition of EUTRA UE Positioning test case 7.3.4.2_6s	10.1.0	10.2.0
2013-06	RAN#60	R5s130288	0013	-	Addition of EUTRA UE Positioning test case 7.3.4.4_5s	10.1.0	10.2.0
2013-06	RAN#60	R5s130290	0014	-	Addition of EUTRA UE Positioning test case 7.3.5.1_1s	10.1.0	10.2.0
2013-06	RAN#60	RP-130616	0015	-	CR to 37.571-4: Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.571-4 (prose), Annex A	10.1.0	10.2.0
2013-09	RAN#61	R5s130421	0016	-	Correction to EUTRA UE Positioning test cases 7.3.4.2.5s and 7.3.4.4.5s	10.2.0	10.3.0
2013-09	RAN#61	R5s130457	0017	-	Correction to EUTRA UE Positioning test case 7.3.5.1.1s	10.2.0	10.3.0
2013-09	RAN#61	R5s130468	0018	-	Correction to EUTRA UE Positioning test cases 7.3.1.1	10.2.0	10.3.0
2013-09	RAN#61	RP-131167	0019	-	CR to 37.571-4: Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.571-4 (prose), Annex A	10.2.0	10.3.0
2013-12	RAN#62	R5s130618	0020	-	Addition of EUTRA UE Positioning test case 7.3.4.1_1s	10.3.0	10.4.0
2013-12	RAN#62	R5s130627	0021	-	Addition of EUTRA UE Positioning test case 7.3.4.3_1s	10.3.0	10.4.0
2013-12	RAN#62	RP-131871	0022	-	CR to 37.571-4: Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.571-4 (prose), Annex A	10.3.0	10.4.0
2014-03	RAN#63	R5-140326	0023	-	A-GNSS ASN.1 reference	10.4.0	10.5.0
2014-03	RAN#63	R5s130913	0024	-	Correction to function fl_TC_7_3_5_1_GNSS_TestBody	10.4.0	10.5.0
2014-03	RAN#63	R5s130914	0025	-	Correction to svID values in AcquisitionAssist and AuxiliaryInformation	10.4.0	10.5.0
2014-03	RAN#63	R5s131009	0026	-	Addition of EUTRA UE Positioning test case 7.3.5.1_2s	10.4.0	10.5.0
2014-03	RAN#63	R5s131011	0027	-	Addition of EUTRA UE Positioning test case 7.3.5.1_4s	10.4.0	10.5.0
2014-03	RAN#63	RP-140316	0028	-	CR to 37.571-4: Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.571-4 (prose), Annex A	10.4.0	10.5.0
2014-06	RAN#64	R5s140086	0029	-	Addition of EUTRA UE Positioning test case 7.2.1.1	10.5.0	10.6.0
2014-06	RAN#64	R5s140088	0030	-	Addition of EUTRA UE Positioning test case 7.2.1.2	10.5.0	10.6.0

Change history

Date	TSG #	TSG Doc.	CR	Rev	Subject/Comment	Old	New
2014-06	RAN#64	R5s140090	0031	-	Addition of EUTRA UE Positioning test case 7.2.1.3	10.5.0	10.6.0
2014-06	RAN#64	R5s140242	0032	-	Correction to LTE UE Positioning test case 7.3.5.1	10.5.0	10.6.0
2014-06	RAN#64	R5s140253	0033	-	Addition of EUTRA UE Positioning test case 7.3.4.4.1s	10.5.0	10.6.0
2014-06	RAN#64	R5s140255	0034	-	Addition of EUTRA UE Positioning test case 7.3.4.2.1s	10.5.0	10.6.0
2014-06	RAN#64	R5s140257	0035	-	Correction to LTE UE Positioning test case 7.3.4.3.1s	10.5.0	10.6.0
2014-06	RAN#64	R5s140316	0036	-	Addition of UTRAN UE Positioning test case 6.2.3.3	10.5.0	10.6.0
2014-06	RAN#64	R5s140318	0037	-	Addition of UTRAN UE Positioning test case 6.2.3.4	10.5.0	10.6.0
2014-06	RAN#64	R5s140320	0038	-	Addition of UTRAN UE Positioning test case 6.2.3.5	10.5.0	10.6.0
2014-06	RAN#64	R5s140433	0039	-	Addition of GCF WI-166 EUTRA UE Positioning test case 7.3.4.3.4s	10.5.0	10.6.0
2014-06	RAN#64	R5s140435	0040	-	Addition of GCF WI-166 EUTRA UE Positioning test case 7.3.4.4.4s	10.5.0	10.6.0
2014-06	RAN#64	RP-140825	0041	-	CR to 37.571-4: Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.571-4 (prose), Annex A	10.5.0	10.6.0
2014-06	RAN#64	R5s140086	0029	-	Addition of EUTRA UE Positioning test case 7.2.1.1	10.5.0	10.6.0
2014-06	RAN#64	R5s140088	0030	-	Addition of EUTRA UE Positioning test case 7.2.1.2	10.5.0	10.6.0
2014-06	RAN#64	R5s140090	0031	-	Addition of EUTRA UE Positioning test case 7.2.1.3	10.5.0	10.6.0
2014-09	RAN#65	R5-144256	0042	-	Routine maintenance and updates	10.6.0	10.7.0
2014-12	RAN#66	R5s140739	0043	-	Addition of GCF WI-166 EUTRA UE Positioning test case 7.3.4.2.4s	10.7.0	10.8.0
2014-12	RAN#66	R5s140777	0044	-	Usage of <MULTIPLE_CNF> in UpperTesterFunctions	10.7.0	10.8.0
2014-12	RAN#66	RP-142067	0045	-	CR to 37.571-4: Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.571-4 (prose), Annex A	10.7.0	10.8.0
2015-03	RAN#67	R5s141359	0047	-	Correction of errors in implementation of positioning CRs	10.8.0	10.9.0
2015-03	RAN#67	R5s150027	0048	-	Addition of GCF WI-166 EUTRA UE Positioning test case 7.3.4.1.4s	10.8.0	10.9.0
2015-03	RAN#67	RP-150336	0049	-	CR to 37.571-4: Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.571-4 (prose), Annex A	10.8.0	10.9.0
2015-03	RAN#67	-	-	-	upgraded to Rel-11 with no change	10.9.0	11.0.0
2015-03	RAN#67	R5-150735	0046	-	Addition of BeiDou Test Model	11.0.0	12.0.0
2015-06	RAN#68	RP-150897	0050	-	CR to 37.571-4: Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.571-4 (prose), Annex A	12.0.0	12.1.0
2015-06	RAN#68	R5s150013	0051	-	Addition of UTRAN UE Positioning test case 6.2.3.2.4s	12.0.0	12.1.0
2015-06	RAN#68	R5s150222	0052	-	Correction to POS ATS in TTCN-3 delivery 15wk10	12.0.0	12.1.0
2015-06	RAN#68	R5s150331	0053	-	Rel-12 baseline upgrade for POS Test Suite	12.0.0	12.1.0
2015-06	RAN#68	R5s150343	0054	-	Corrections to UTRAN UE Positioning functions	12.0.0	12.1.0
2015-06	RAN#68	R5s150344	0055	-	Correction to LTE UE Positioning test cases 7.3.5.1.5s and 7.5.1	12.0.0	12.1.0
2015-06	RAN#68	R5s150345	0056	-	Correction to GLONASS LPP test cases	12.0.0	12.1.0
2015-06	RAN#68	R5s150353	0057	-	Improvement of function f_POS_CheckGnssSupportList	12.0.0	12.1.0

Change history

Date	TSG #	TSG Doc.	CR	Rev	Subject/Comment	Old	New
2015-06	RAN#68	R5s150404	0058	-	Addition of GCF WI-166 EUTRA UE Positioning test case 7.3.3.1	12.0.0	12.1.0
2015-06	RAN#68	R5s150439	0059	-	Correction to selection of Assistance Data for sub-test 4	12.0.0	12.1.0
2015-09	RAN#69	RP-151421	0060	-	CR to 37.571-4: Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.571-4 (prose), Annex A	12.1.0	12.2.0
2015-09	RAN#69	R5s150011	0061	-	Addition of UTRAN UE Positioning test case 6.2.1.1.4s	12.1.0	12.2.0
2015-09	RAN#69	R5s150589	0062	-	Correction to EUTRA Positioning functions	12.1.0	12.2.0
2015-09	RAN#69	R5s150590	0063	-	Correction to Upper Tester modules	12.1.0	12.2.0
2015-09	RAN#69	R5s150706	0064	-	Addition of UTRAN UE Positioning test case 6.2.2.1.4s	12.1.0	12.2.0
2015-09	RAN#69	R5s150708	0065	-	Addition of UTRAN UE Positioning test case 6.2.2.3.4s	12.1.0	12.2.0
2015-09	RAN#69	-	-	-	update of the "non-specific references" in section 2 according to the approved R5-153582 and an action point on ETSI MCC	12.1.0	12.2.0
2015-12	RAN#70	R5s150710	0068	-	Addition of GCF WI-166 EUTRA UE Positioning test case 7.3.4.2.7s	12.2.0	12.3.0
2015-12	RAN#70	R5s150712	0069	-	Addition of GCF WI-166 EUTRA UE Positioning test case 7.3.4.4.7s	12.2.0	12.3.0
2015-12	RAN#70	R5s150805	0070	-	Correction to POS capability check procedure	12.2.0	12.3.0
2015-12	RAN#70	R5s150911	0071	-	Code improvements in Positioning test cases	12.2.0	12.3.0
2015-12	RAN#70	RP-151697	0072	-	CR to 37.571-4: Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.571-4 (prose), Annex A	12.2.0	12.3.0
2016-03	RAN#71	R5s151010	0073	-	Correction to POS type definition in ATS_15wk38	12.3.0	12.4.0
2016-06	RAN#72	R5s160479	0074	1	Correction to LTE Positioning test cases	12.4.0	12.5.0
2016-06	RAN#72	R5s160480	0075	1	Correction to UTRA Positioning test cases	12.4.0	12.5.0
2016-09	RAN#73	R5s160713	0076	-	Rel-13 baseline upgrade for POS Test Suite	12.5.0	13.0.0
2016-12	RAN#74	-	-	-	updated to deliver latest TTCN	13.0.0	13.1.0
2017-03	RAN#75	R5-171531	0082	-	Indoor Positioning: Introduction of Test Model	13.1.0	13.2.0
2017-03	RAN#75	R5s170127	0078	1	Correction to GCF WI-166 EUTRA UE Positioning test case 7.3.4.4.7s	13.1.0	13.2.0
2017-03	RAN#75	R5s170128	0079	1	Correction to function f_POS_CheckCapabilities() for OTDOA	13.1.0	13.2.0
2017-03	RAN#75	R5s170186/RP-170104	0084		Update ASP changes due to inclusion of iPOS in 37.571-4 (prose), Annex D	13.1.0	13.2.0
2017-06	RAN#76	R5s170311	0085	-	Addition of LTE UE Positioning test case 7.3.4.1.10s	13.2.0	13.3.0
2017-06	RAN#76	R5s170313	0086	-	Addition of LTE UE Positioning test case 7.3.4.3.10s	13.2.0	13.3.0
2017-06	RAN#76	R5s170363	0087	-	Aligning name of PICS used in TTCN to the TS 37.571-3 definition	13.2.0	13.3.0
2017-06	RAN#76	R5s170383	0083	1	Corrections to UE Positioning test cases	13.2.0	13.3.0
2017-06	RAN#76	R5s170416	0088	-	Update of TC list in 37.571-4 (prose), Annex A, due to merging of GNSS subtest	13.2.0	13.3.0
2017-09	RAN#77	R5s170630	0089	1	Correction to function 'f_POS_CheckGnssSupportList'	13.3.0	13.4.0

Change history

Date	TSG #	TSG Doc.	CR	Rev	Subject/Comment	Old	New
2017-09	RAN#77	R5s170599	0090		Rel-14 baseline upgrade for POS Test Suite	13.4.0	14.0.0
2017-09	RAN#77	R5-173663	0091	-	Indoor Positioning Enhancements: Introduction of Test Model	13.4.0	14.0.0
2017-12	RAN#78	R5-176982	0093	1	GNSS Test Model updates for V2X	14.0.0	14.1.0
2017-12	RAN#78	R5s170860	0092	1	Correction to UE Positioning test case 7.3.1.1	14.0.0	14.1.0
2018-03	RAN#79	R5-180677	0099	-	Routine maintenance for TS 37.571-4	14.1.0	14.2.0
2018-03	RAN#79	R5s180081	0094	1	Correction to EUTRA release information for UE Positioning TCs	14.1.0	14.2.0
2018-03	RAN#79	R5s180113	0095	1	Addition of LTE UE Positioning test case 7.3.4.2.15s (BDS)	14.1.0	14.2.0
2018-03	RAN#79	R5s180114	0096	1	Addition of LTE UE Positioning test case 7.3.4.4.15s (BDS)	14.1.0	14.2.0
2018-03	RAN#79	R5s180115	0097	1	Addition of LTE UE Positioning test case 7.3.5.1.15s (BDS)	14.1.0	14.2.0
2018-06	RAN#80	R5-182292	0102	-	V2X: Test Model updates	14.2.0	14.3.0
2018-06	RAN#80	R5s180327	0101	1	Correction / improvement to UE Positioning TCs 7.2.2.1 + 7.2.2.2	14.2.0	14.3.0
2018-12	RAN#82	R5s180569	0105	-	Correction to number of almanac elements for Galileo	14.3.0	14.4.0
2018-12	RAN#82	R5s180554	0103	-	Rel-15 Sep'18 baseline upgrade for POS Test Suite	14.4.0	15.0.0
2019-03	RAN#83	R5s190056	0106	-	Correction to checking LPP ePDU capabilities	15.0.0	15.1.0
2019-09	RAN#85	R5-197174	0111	1	UE Positioning over NR: Initial Test Model	15.1.0	15.2.0
2019-09	RAN#85	R5s190462	0109	-	Correction to f_POS_CheckCapabilities_Agnss	15.1.0	15.2.0
2019-09	RAN#85	R5s190463	0110	-	Correction to f_POS_CheckGnssSupportList	15.1.0	15.2.0
2019-09	RAN#85	R5s190759	0112	-	Rel-15 Jun'19 partial baseline upgrade for POS TTCN-3 Test Suite	15.1.0	15.2.0
2019-12	RAN#86	R5-198140	0113	-	Aerial Vehicles: Test Model addition	15.2.0	15.3.0
2019-12	RAN#86	R5-198142	0114	-	UE Positioning over NR: Test Model updates	15.2.0	15.3.0
2019-12	RAN#86	R5-198156	0115	-	Routine maintenance for TS 37.571-4	15.2.0	15.3.0
2019-12	RAN#86	R5s191048	0116	-	Correction to the initialisation function for LTE Positioning test cases	15.2.0	15.3.0
2020-03	RAN#87	R5-200261	0118	-	Routine maintenance for TS 37.571-4	15.3.0	15.4.0
2020-03	RAN#87	R5-200262	0119	-	Editorial changes to TS 37.571-X titles to remove references to individual RATs	15.3.0	15.4.0
2020-06	RAN#88	R5s200581	0120	-	Correction to f_POS_CheckGnssSupportList	15.4.0	15.5.0
2020-12	RAN#90	R5s201391	0122	-	Addition of 5G UE Positioning test case 9.3.1.1	15.5.0	15.6.0
2020-12	RAN#90	R5s201393	0123	-	Addition of 5G UE Positioning test case 9.3.1.2.15s	15.5.0	15.6.0
2020-12	RAN#90	R5s201395	0124	-	Addition of 5G UE Positioning test case 9.3.2.1	15.5.0	15.6.0
2020-12	RAN#90	R5s201397	0125	-	Addition of 5G UE Positioning test case 9.3.2.2	15.5.0	15.6.0
2020-12	RAN#90	R5s201399	0126	-	Addition of 5G UE Positioning test case 9.3.2.3	15.5.0	15.6.0
2020-12	RAN#90	R5s201401	0127	-	Addition of 5G UE Positioning test case 9.3.4.1.15s	15.5.0	15.6.0
2020-12	RAN#90	R5s201403	0128	-	Addition of 5G UE Positioning test case 9.3.4.2.15s	15.5.0	15.6.0

Change history

Date	TSG #	TSG Doc.	CR	Rev	Subject/Comment	Old	New
2020-12	RAN#90	R5s201405	0129	-	Addition of 5G UE Positioning test case 9.3.4.3.15s	15.5.0	15.6.0
2020-12	RAN#90	R5s201407	0130	-	Addition of 5G UE Positioning test case 9.3.4.4.15s	15.5.0	15.6.0
2020-12	RAN#90	R5s201460	0131	-	Correction to several NR5GC UE Positioning Test related functions	15.5.0	15.6.0
2020-12	RAN#90	R5s201529	0132	-	Correction to function f_POS_NR_SendLppAck()	15.5.0	15.6.0
2020-12	RAN#90	R5s201649	0134	-	Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.571-4 (prose), Annex A	15.5.0	15.6.0
2020-12	RAN#90	R5s201388	0121	-	Rel-16 Sep'20 baseline upgrade for POS Test Suite	15.6.0	16.0.0
2021-03	RAN#91	R5s210208	0133	1	Correction to function f_POS_NR_CheckAgnssMultiFreq	16.0.0	16.1.0
2021-03	RAN#91	R5s210262	0135	1	Correction to a NR5GC UE Positioning related function	16.0.0	16.1.0
2021-06	RAN#92	R5-212285	0138	-	Routine maintenance for TS 37.571-4	16.1.0	16.2.0
2021-06	RAN#92	R5s210416	0136	-	Correction for common POS function f_POS_SelectTemplateReq_IonosphModel()	16.1.0	16.2.0
2021-09	RAN#93	R5-214729	0143	-	Routine maintenance for TS 37.571-4	16.2.0	16.3.0
2021-09	RAN#93	R5s210887	0144	-	Correction to NR5GC Positioning Functions	16.2.0	16.3.0
2021-09	RAN#93	R5s210922	0141	1	Correction for multi frequency GNSS support checking	16.2.0	16.3.0
2021-09	RAN#93	R5s210926	0142	1	Addition of NR5GC Positioning test case 9.3.1.2.5s	16.2.0	16.3.0
2021-09	RAN#93	R5s211177	0145	-	Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.571-4 (prose), Annex A	16.2.0	16.3.0
2021-12	RAN#94	R5-216658	0149	-	NR Positioning: Test Model addition	16.3.0	16.4.0
2021-12	RAN#94	R5-217256	0151	-	Include new pixit for the 2012 GNSS scenarios	16.3.0	16.4.0
2021-12	RAN#94	R5s211307	0147	-	Correction to POS_Parameters	16.3.0	16.4.0
2021-12	RAN#94	R5s211515	0148	-	Correction to f_POS_OTDOA_CheckEutraBand	16.3.0	16.4.0
2021-12	RAN#94	R5s211696	0153	-	Add new ASP definitions for NR test model extension in Rel-16	16.3.0	16.4.0
2022-03	RAN#95	R5-220457	0156	-	5G V2X: GNSS Test Model updates for NR sidelink	16.4.0	16.5.0
2022-03	RAN#95	R5s220170	0152	1	Correction to NR5GC_Positioning_Functions for capability check	16.4.0	16.5.0
2022-03	RAN#95	R5s220432	0154	1	Correction to f_POS_NR_CheckCapabilities_NR_Ecid	16.4.0	16.5.0
2022-03	RAN#95	R5s220433	0155	1	Correction to SVIDs for 2012 GNSS scenario	16.4.0	16.5.0
2022-03	RAN#95	R5s220439	0158	-	Update of TTCN type definitions, Annex D	16.4.0	16.5.0
2022-06	RAN#96	R5s220671	0159	-	Correction to f_UT_AT_CheckNotifyLocationRequest	16.5.0	16.6.0
2022-06	RAN#96	R5s220672	0160	-	Correction to f_POS_NR_Set_DlPrsPeriodicityAndOffset_11ms	16.5.0	16.6.0
2022-09	RAN#97	R5s220765	0163	-	Correction to function f_TC_7_3_4_2_7s_EUTRA	16.6.0	16.7.0
2022-09	RAN#97	R5s220799	0164	-	Correction to f_TC_7_3_4_4_7s_EUTRA	16.6.0	16.7.0
2022-09	RAN#97	R5s220900	0162	1	Addition of LTE POS testcase 7.2.2.2.5s	16.6.0	16.7.0
2022-09	RAN#97	R5s221026	0165	-	Rel-16 Mar'22 partial baseline upgrade for POS Test Suite	16.6.0	16.7.0

Change history

Date	TSG #	TSG Doc.	CR	Rev	Subject/Comment	Old	New
2022-09	RAN#97	R5s221073	0173	-	Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.571-4 (prose), Annex A	16.6.0	16.7.0
2022-12	RAN#98	R5s221296	0166	1	Correction to f_POS_NR_CheckCapabilities_Prs_ProcessingCapabilityBandList	16.7.0	16.8.0
2022-12	RAN#98	R5s221350	0168	1	Addition of POS testcase 9.3.1.2.19s	16.7.0	16.8.0
2022-12	RAN#98	R5s221351	0169	1	Addition of POS testcase 9.3.4.2.19s	16.7.0	16.8.0
2022-12	RAN#98	R5s221352	0170	1	Addition of POS testcase 9.3.4.2.20s	16.7.0	16.8.0
2022-12	RAN#98	R5s221353	0171	1	Addition of POS testcase 9.3.4.2.21s	16.7.0	16.8.0
2022-12	RAN#98	R5s221354	0172	1	Addition of POS testcase 9.3.4.2.22s	16.7.0	16.8.0
2022-12	RAN#98	R5s221363	0175	-	Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.571-4 (prose), Annex A	16.7.0	16.8.0
2022-12	RAN#98	R5s221180	0174	-	Rel-17 Sep'22 baseline upgrade for POS Test Suite	16.8.0	17.0.0
2023-03	RAN#99	R5s230101	0178	-	Addition of NR POS testcase 9.3.4.4.21s	17.0.0	17.1.0
2023-03	RAN#99	R5s230205	0176	1	Addition of NR POS testcase 9.3.4.4.19s	17.0.0	17.1.0
2023-03	RAN#99	R5s230206	0177	1	Addition of NR POS testcase 9.3.4.4.20s	17.0.0	17.1.0
2023-03	RAN#99	R5s230260	0179	-	Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.571-4 (prose), Annex A	17.0.0	17.1.0
2023-06	RAN#100	R5s230438	0180	-	Addition of LTE POS testcase 7.2.2.2.6.FDDs	17.1.0	17.2.0
2023-06	RAN#100	R5s230440	0181	-	Addition of LTE POS testcase 7.2.2.2.6.TDDs	17.1.0	17.2.0
2023-06	RAN#100	R5s230473	0182	-	Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.571-4 (prose), Annex A	17.1.0	17.2.0
2023-09	RAN#101	R5s230498	0183	-	Correction for NR5GC positioning test cases	17.2.0	17.3.0
2023-09	RAN#101	R5s230499	0184	-	Correction for EUTRA positioning test case 7.3.4.4_7s	17.2.0	17.3.0
2023-09	RAN#101	R5s230502	0185	-	Addition of CNF templates in PosSystem_Control	17.2.0	17.3.0
2023-09	RAN#101	R5s230513	0186	-	Correction to f_NR_ConfigUlSrsMultiRTT	17.2.0	17.3.0
2023-09	RAN#101	R5s230595	0187	-	Correction for GNSS 15s test cases	17.2.0	17.3.0
2023-09	RAN#101	R5s230608	0188	-	Correction to svID values in AcquisitionAssist and AuxiliaryInformation for GNSS Scenarios 2020	17.2.0	17.3.0
2023-09	RAN#101	R5s230674	0191	-	Update of TTCN type definitions, Annexes C and E	17.2.0	17.3.0
2023-12	RAN#102	R5s230711	0192	-	Correction for positioning test cases	17.3.0	17.4.0
2023-12	RAN#102	R5s230755	0193	-	Correction for function f_POS_NR_CheckCapabilities_Prs_ProcessingCapabilityBandList()	17.3.0	17.4.0
2023-12	RAN#102	R5s230842	0189	1	Rel-17 June'23 partial baseline upgrade for POS Test Suite	17.3.0	17.4.0
2023-12	RAN#102	R5s230846	0190	1	Corrections on control module for POS Test Suite	17.3.0	17.4.0
2024-03	RAN#103	R5-240553	0198	-	Removal of PIXIT for deprecated signalling GNSS scenarios	17.4.0	17.5.0
2024-03	RAN#103	R5s240144	0195	-	Correction to f_POS_NR_Set_DlPrsAssistanceDataPerTRP	17.4.0	17.5.0
2024-03	RAN#103	R5s240146	0196	-	Addition of NR5GC POS test case 9.3.1.2.21s	17.4.0	17.5.0
2024-03	RAN#103	R5s240160	0197	-	Correction to f_POS_NR_CheckCapabilities	17.4.0	17.5.0

Change history

Date	TSG #	TSG Doc.	CR	Rev	Subject/Comment	Old	New
2024-03	RAN#103	R5s240169	0200	-	Addition of NR POS Test case 9.3.1.2.20s	17.4.0	17.5.0
2024-03	RAN#103	R5s240199	0194	1	Correction to function f_POS_NR_CheckCapabilities_NR_Ecid	17.4.0	17.5.0
2024-03	RAN#103	R5s240240	0201	-	Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.571-4 (prose), Annex A	17.4.0	17.5.0
2024-06	RAN#104	R5s240301	0202	-	Correction of GNSS 15s test cases for BDS	17.5.0	17.6.0
2024-09	RAN#105	R5s240532	0205	-	Rel-18 June'24 baseline upgrade for POS Test Suite	17.6.0	18.0.0
2025-03	RAN#107	R5s250032	0207	-	Rel-18 Dec'24 partial baseline upgrade for POS Test Suite	18.0.0	18.1.0
2025-09	RAN#109	R5-253976	0209	-	NR POS enh. Phase2: Initial Test Model	18.1.0	18.2.0
2025-09	RAN#109	R5s250329	0208	-	Correction to function f_POS_NR_CheckCapabilities_Prs_ProcessingCapabilityBandList	18.1.0	18.2.0
2025-09	RAN#109	R5s250343	0210	-	Correction to function f_POS_NR_CheckCapabilities_Srs_PosResourceConfigCA_BandList	18.1.0	18.2.0
2025-12	RAN#110	R5-255777	0211	-	NR UAV: Initial Test Model	18.2.0	18.3.0
2026-03	RAN#111	R5-260283	0213	-	Routine maintenance for TS 37.571-4	18.3.0	18.4.0
2026-03	RAN#111	R5-261183	0212	1	NR POS enh. Phase2: Test Model updates	18.3.0	18.4.0
2026-03	RAN#111	R5s260075	0214	1	Update of TTCN type definitions, Annex F	18.3.0	18.4.0
2026-06	RAN#112	R5-261930	0217		NR POS enh. Phase2: Test Model updates	18.4.0	18.5.0
2026-06	RAN#112	R5s260173	0218		Correction to tsc_NoOfCordPorts to accommodate new Sidelink PTC	18.4.0	18.5.0
2026-06	RAN#112	R5s260106	0216		Rel-19 Mar'26 baseline upgrade for POS Test Suite	18.5.0	19.0.0

History

Version	Date	Status
V19.0.0	August 2026	Publication